

GENERAL SCIENCE GRADE 10: WAVES

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 3.3 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	General Science
Strand	Strand 3.0: Natural Physical Science
Sub-Strand	Sub-Strand 3.3: Waves
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Terms used in waves (wavelength, period, amplitude, frequency and velocity) – The wave equation ($v = f\lambda$) – Properties of waves (reflection, refraction, diffraction) – Application of properties of waves in real life
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - explain the terms used in waves, - interpret the wave equation as used in science, - calculate the characteristics using the wave equation, - demonstrate the effects of properties of waves in real life, - appreciate the applications of waves in real life.
Core Competencies	– Learning to learn: as the learner searches for information on applications of reflection, refraction and diffraction of waves and reflects on findings from experiments and peer discussions. – Critical thinking and problem solving: as the learner searches for information on mitigation measures of the effects of waves on the environment (communities, ecosystems, infrastructure and operations) and solves numerical problems using the wave equation.
Core Values	– Unity: as the learners play different roles when carrying out experiments to investigate reflection of sound waves using the echo method.
Science & Engineering Practices	– Asking questions and defining problems: as learners pose investigable questions about wave phenomena observed at Lake Victoria, Mombasa coastline, and Nairobi's busy highways. – Planning and carrying out investigations: as learners design and conduct experiments to investigate reflection of sound waves using the echo method and explore diffraction of sound around corners. – Analysing and interpreting data: as learners interpret oscilloscope traces, wave diagrams, and experimental results to identify wavelength, amplitude, frequency, and period. – Using mathematics and computational thinking: as learners solve numerical problems using the wave equation ($v = f\lambda$) to calculate wave speed, frequency, and wavelength. – Constructing explanations: as learners explain real-life wave phenomena — including why a matatu driver's radio fades in the

	Rift Valley tunnels and why fishermen on Lake Victoria hear distant thunderclaps before others. – Obtaining, evaluating, and communicating information: as learners research effects of waves on marine ecosystems and infrastructure, then discuss mitigation measures with peers.
Pertinent & Contemporary Issues (PCIs)	– Safety and security: as the learner appreciates some of the applications of waves related to safety in transport. – Environmental issues: as the learner searches for information on the effect of waves on the environment (communities, marine ecosystems, infrastructure and operations).
Career Connections	– Telecommunications engineer: designing radio, TV, and mobile-network systems that transmit signals as electromagnetic waves across Kenya's communication infrastructure. – Marine/fisheries technologist: applying knowledge of water-wave behaviour to protect fishing communities on Lake Victoria and coastal Mombasa from wave hazards. – Medical imaging technician: using ultrasound wave principles in Kenyan hospitals for diagnostic scanning (e.g., Kenyatta National Hospital). – Civil/structural engineer: assessing and mitigating the destructive effects of seismic and water waves on roads, bridges, and buildings across Kenya. – Meteorologist: applying wave properties to interpret weather radar and satellite data for forecasting rainfall patterns in the Rift Valley and coastal regions. – Road-safety officer: leveraging reflection and refraction principles in the design of road signage, cats-eye reflectors, and speed-camera systems on Kenyan highways.
Focus for Lessons	This unit investigates the nature, properties, and real-life applications of waves, anchored in the observable and puzzling behaviour of sound and water waves within Kenyan everyday contexts — from the echo heard across the walls of Nairobi's tall buildings to the way waves batter the Mombasa coastline. Learners progress from building a precise wave vocabulary and mastering the wave equation, through hands-on experiments on reflection, refraction, and diffraction, to critically evaluating the impact of waves on Kenya's communities, marine ecosystems, and infrastructure. Throughout, the storyline is driven by the question of how invisible energy travelling as waves shapes the safety, technology, and environment of daily Kenyan life.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How do waves carry energy through our environment — and how can understanding their behaviour protect Kenyan communities, improve technology, and help us make sense of the world around us? KICD KEY INQUIRY QUESTIONS: 1. How are waves applicable in our day-to-day life? 2. How do waves affect our environment?
Anchoring Phenomenon	A fisherman on the shores of Lake Victoria, near Kisumu, notices something puzzling during a fierce evening storm: he can hear the deep rumble of thunder from a distant lightning strike long after he sees the flash, and the sound seems to bend around the rocky headland so that it reaches him even when the storm is clearly behind the cliff. At the same time, the water waves radiating outward from the centre of the storm change direction as they cross from the deep open lake into the shallower bay near the shore, and they seem to spread around the headland and into the protected cove — even though there is no straight-line path from the storm to that cove. Students are asked: how can sound reach a place it should not be able to reach, why does water change direction at the shallow edge, and what does any of this have to do with the mobile-phone signal that fades when you go behind a hill in the Rift Valley? This multi-sensory, locally grounded phenomenon is puzzling because learners have everyday experience of sound and water but cannot yet explain the bending, bouncing, and spreading of wave energy using scientific language or mathematical relationships.
Storyline Thread	Lesson 1: ANCHORING PHENOMENON & WAVE VOCABULARY — Students encounter the Lake Victoria fisherman scenario

through a short narrated audio clip and a photograph of the Kisumu shoreline during a storm. In the Predict phase, learners write individual predictions explaining why the fisherman hears thunder from behind a cliff. Class shares predictions; teacher records them on the Driving Question Board (DQB). Learners then use print or non-print media to research wave terminology (wavelength, period, amplitude, frequency, velocity) and construct a labelled wave diagram with Kenyan-context annotations (e.g., marking the wavelength of a Lake Victoria water wave versus a spoken Swahili vowel sound). Lesson closes with learners adding their first "What I wonder..." sticky notes to the DQB.

Lesson 2: THE WAVE EQUATION — Building on vocabulary, learners Observe a teacher demonstration using a slinky spring and a ripple tank (or simulation) to show that a faster-vibrating source produces shorter-wavelength waves. Students measure period and frequency from the demonstration, then derive the relationship $v = f\lambda$ through guided questioning (WAIT TIME: 30 seconds after each prompt; cold-call minimum 4 students per key step). Explain phase: learners solve a set of four numerical problems contextualised in Kenya — calculating the wavelength of a Nairobi FM radio station (e.g., 98.4 MHz), the speed of sound during a Mombasa echo, the frequency of a fisherman's boat-engine vibration on Lake Victoria, and the period of a coastal Indian Ocean swell. Learners update the DQB with new understanding and refine their anchoring-phenomenon predictions.

Lesson 3: REFLECTION OF WAVES — Predict phase: learners predict what will happen to a torch-beam (or sound pulse) when it hits a hard flat wall versus a curved wall. Observe phase: learners carry out the echo experiment — standing measured distances from the school's perimeter wall, clapping rhythmically and timing echoes to calculate sound speed; results are recorded and compared across groups. Explain phase: learners draw incident and reflected ray diagrams for sound at the Nairobi building walls, connecting back to the fisherman hearing sound from "behind" the cliff. Teacher introduces the law of reflection using protractor measurements from the experiment. DQB updated: "Reflection explains part of the fisherman's puzzle — but does it explain the sound going AROUND the headland?"

Lesson 4: REFRACTION OF WAVES — Predict phase: learners predict whether sound travels faster in warm air (above hot ugali on a jiko) or cool air and what happens to its direction. Observe phase: learners watch a video clip showing water waves bending as they cross from deep to shallow water in a ripple tank, with a Mombasa coastal-swell footage clip as a real-life parallel. They illustrate the concept diagrammatically (incident ray, refracted ray, normal, boundary). Explain phase: learners discuss in groups why the Lamu fishermen's calls travel farther at night (temperature inversion refracts sound downward), and connect this to the matatu FM-radio fading in the Rift Valley. Numerical problem: given wave speed changes from 340 m/s to 320 m/s at an air-temperature boundary, calculate the new wavelength if frequency is constant. DQB updated with the refraction explanation for shallow-water bending.

Lesson 5: DIFFRACTION OF WAVES — Predict phase: learners predict whether they can hear a person speaking on the other side of a doorway without seeing them, and why. Observe phase: learners carry out the experiment establishing that sound waves can be heard round a corner — one group claps or speaks near an open classroom doorway while others listen around the corner at increasing distances; results are recorded. They also watch a simulation of water waves diffracting through a gap of varying width. Explain phase: learners construct ray/wavefront diagrams showing diffraction through a narrow gap (representing the rocky headland cove from the anchoring phenomenon) and a wide gap. Learners connect diffraction to mobile-phone signals bending around hills in the Rift Valley and to the spread of radio waves between Nairobi and Kisumu. DQB updated: the full anchoring phenomenon (echo = reflection; shallow-bay bending = refraction; sound around the headland = diffraction) is now explained in learner-generated notes.

Lesson 6: APPLICATIONS OF WAVE PROPERTIES IN REAL LIFE — Learners move into an Explain/Apply phase. Working in groups, they research one assigned application each: (a) SONAR for depth-sounding by Kenya Marine and Fisheries Research Institute on Lake Victoria; (b) ultrasound scanning at Kenyatta National Hospital; (c) cats-eye road reflectors and speed cameras

	on the Nairobi–Mombasa highway; (d) earthquake (seismic) wave monitoring by the Kenya Meteorological Department. Groups create a poster or digital slide and present to the class (3 minutes each, followed by 2 minutes of peer questions). Teacher facilitates a whole-class discussion linking each application back to reflection, refraction, or diffraction. DQB: learners add application cards beside the corresponding wave property. Lesson 7: EFFECTS OF WAVES ON THE ENVIRONMENT & MITIGATION — Observe/Research phase: learners use print or non-print media to search for information on the effect of waves on the environment — coastal erosion at Mombasa and Lamu, impact of seismic waves on Nairobi's infrastructure, noise pollution from wave energy in urban areas, and disruption of marine ecosystems in Lake Victoria. Explain phase: learners discuss mitigation measures (sea walls, building codes, noise barriers, marine protected areas) with peers and record findings. A structured debate (teams of 3) addresses: "Should Kenya invest in a coastal wave-energy power station? What are the environmental trade-offs?" DQB updated with environmental impact and mitigation cards. Learners begin drafting their cumulative model. Lesson 8: MODEL BUILDING, NUMERICAL CONSOLIDATION & DQB CLOSURE — Learners individually complete a final set of mixed numerical problems (wave equation, echo-distance calculation, wavelength shift on refraction) using the wave equation in contexts drawn from across the unit's Kenyan scenarios. Then, in pairs, they produce or refine their cumulative annotated diagram (Model Building phase) showing: the Lake Victoria anchoring phenomenon with labelled reflection (echo off cliff), refraction (shallow bay), and diffraction (sound around headland); the wave equation linking their measurements; and at least two real-life applications and one environmental impact with a mitigation measure. Class reconvenes for DQB Closure — teacher reads each original "I wonder..." question aloud; cold-calls a student to answer using evidence from the unit (minimum 6 students called). Unanswered questions are flagged for further inquiry. Unit closes with learners writing a one-paragraph "Scientist's Statement" explaining to a Kisumu fisherman how wave science keeps coastal communities safer.
Supporting Phenomena	– A student clapping once between two parallel walls of a school building in Nairobi hears multiple distinct echoes — a clear, repeatable instance of sound-wave reflection that raises the question of what determines the time gap between echoes and the wave equation link to distance and speed. – Matatu passengers travelling through the Rift Valley escarpment road notice that FM radio reception cuts in and out as the vehicle moves behind hills or into cuttings, illustrating diffraction and absorption of electromagnetic waves in a familiar transport context. – During a Geography field trip to Mombasa, learners observe that ocean swells arriving from the open Indian Ocean slow down, shorten in wavelength, and bend toward the shore as the sea becomes shallower — a visible, real-time demonstration of wave refraction in water that parallels refraction of sound and light. – Fishermen in Lamu use shouted calls that travel far over calm water at night but seem to disappear in the daytime bustle — a phenomenon explained by the refraction of sound waves as temperature layers in the air change from night to day, linking wave properties directly to the livelihoods of coastal Kenyan communities.

LESSON 1 (40 minutes): ANCHORING PHENOMENON AND WAVE VOCABULARY

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the sub-strand investigation by grounding learners in a locally meaningful puzzling event and equipping them with the vocabulary needed to describe wave behaviour throughout the unit
Knowledge	Learners will identify and define the key wave properties: wavelength, period, amplitude, frequency, and wave velocity, and will distinguish between transverse and longitudinal wave types using diagrams labelled with Kenyan-context examples
Skills	Learners will construct a labelled wave diagram, annotate it with real Kenyan wave examples (Lake Victoria water waves and spoken Swahili vowel sounds), and use print or non-print media to research wave terminology independently
Attitudes	Learners will demonstrate curiosity about natural phenomena in their own environment and show willingness to revise initial ideas as new vocabulary and evidence emerge
Key Inquiry Question	How are waves applicable in our day-to-day life?
Purpose in Storyline	This is the opening lesson that anchors the entire unit. Learners encounter the Lake Victoria fisherman scenario for the first time, generate their initial (pre-instruction) predictions, and build the vocabulary toolkit they will need for every subsequent lesson. The Driving Question Board opened here will grow lesson by lesson until it is closed in Lesson 8.
Safety Notes	

B. LESSON OVERVIEW

Learners are introduced to the anchoring phenomenon of a Kisumu fisherman who hears thunder from behind a cliff and sees water waves bending near the shore. They listen to a short narrated audio clip and examine a photograph of the Kisumu shoreline. They write individual predictions, share them with the class, and the teacher opens the Driving Question Board. Learners then research wave vocabulary using available resources and create a labelled wave diagram with Kenyan-context annotations. The lesson closes with learners posting their first sticky-note questions on the DQB.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Wave properties Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Learners listen to a one-to-two-minute narrated audio clip describing the Lake Victoria fisherman scenario (thunder heard after the lightning flash, sound seemingly coming from behind a rocky cliff, water waves bending near the shore). They simultaneously examine a	Before playing the audio, the teacher says: BEFORE I play this clip, I want you to look at this photograph of the shores of Lake Victoria near Kisumu. Notice the rocky headland, the shallow bay, and the open lake. Keep that image in your mind. The teacher plays the audio clip, then says:	Elicit-then-compare predictions strategy: by recording all predictions publicly on the DQB without evaluation, the teacher surfaces prior conceptions (for example, sound bouncing versus sound bending versus sound travelling through the cliff) and creates productive cognitive	The teacher scans the written predictions as learners write (circulating quietly) and uses cold-call responses to gauge the range of prior knowledge. Look for: learners who already use words like echo or reflection (indicates prior knowledge to build on); learners who attribute the bending

 HTML: Landforms from Wave Erosion and Deposition (Flexbook) Source: CK-12  Search ARES for similar readings	projected or printed photograph of the Kisumu shoreline during stormy weather. Each learner then writes, individually and in silence, their personal prediction in their exercise book, answering the prompt: WHY can the fisherman hear thunder from behind the cliff, and why do the water waves change direction near the shore? Learners are given a full three minutes of silent writing time before any sharing begins. After writing, six to eight learners are cold-called to share their predictions aloud. The teacher records each prediction verbatim on the DQB using a different colour for each prediction category (sound-related, water-related, energy-related, other).	Now write in your book: explain in your own words why you think the fisherman can hear the thunder even though the cliff is in the way, and why the water waves seem to change direction. You have three minutes. Write alone. No talking yet. [WAIT TIME: 3 minutes of silent individual writing.] After writing, the teacher says: I am going to call on some of you. I want to hear YOUR thinking, not a neighbour's. [Cold-calls minimum six students, distributing across the room.] After each response the teacher probes with: What in your everyday experience makes you think that? or Can anyone add to what [name] has said without repeating it? The teacher does NOT confirm or reject any prediction at this stage, responding instead with: That is an interesting idea. Let us put it on our board and see what the science tells us by the end of this unit.	conflict. Learners can see that the class holds multiple competing ideas, which motivates the need for scientific vocabulary and investigation.	entirely to wind (a common misconception to address in Lesson 4 on refraction); learners who express no explanation at all (will need additional scaffolding in the Observe phase). Record two or three notable prediction types in the teacher observation journal.
Observe Phase  VIDEO: Wave properties Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Landforms from Wave Erosion and Deposition (Flexbook) Source: CK-12  Search ARES for similar readings	Learners observe two visual inputs. First, the teacher displays a second image: a labelled diagram projected on the board showing a basic transverse wave and a basic longitudinal wave, with parts unlabelled. Learners do NOT receive labels yet. They observe the diagram for 60 seconds and in pairs discuss: What do you notice? What is the same and what is different between the two wave shapes? Each pair shares one observation. Second, the teacher briefly demonstrates a slinky spring on the floor (or uses a 90-second simulation video if a slinky is unavailable), showing one transverse pulse (side-to-side flick) and one longitudinal pulse (push-pull compression). Learners	As learners observe the unlabelled diagram the teacher says: Look carefully at these two wave patterns. In 60 seconds, tell your partner: what do you notice about the shape, and what is different about how the disturbance moves? [WAIT TIME: 60 seconds.] After pairs share, the teacher says: You have described what you see. Scientists have specific names for every feature you just pointed at. That is what you are going to discover now. During the slinky demonstration the teacher says: Watch this carefully. I am not explaining yet. Just observe. [Performs transverse pulse.] What did you see? [WAIT TIME: 30 seconds. Cold-calls two students.] [Performs longitudinal pulse.] And	Notice-and-wonder observation protocol: by withholding labels during observation, the teacher trains learners to describe physical phenomena in their own language first. This builds observational science skills and ensures that when vocabulary is introduced in the Explain phase, it is anchored to a concrete visual and physical referent rather than an abstract definition.	Listen to pair-share conversations for evidence that learners can distinguish the direction of particle motion from the direction of wave travel. A learner who says the slinky moves side to side but the wave moves forward is demonstrating the conceptual core of transverse waves. Use this as a benchmark: if fewer than half the class articulates this distinction, slow down in the Explain phase and add a second slinky demonstration.

	observe silently and sketch what they see in their books, noting the direction of the disturbance relative to the direction of travel.	this one? How is it different? [WAIT TIME: 30 seconds. Cold-calls two different students.] The teacher does not provide the vocabulary during the Observe phase, preserving learner discovery for the Explain phase.		
Explain Phase  VIDEO: Wave properties Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Landforms from Wave Erosion and Deposition (Flexbook) Source: CK-12  Search ARES for similar readings	Learners use available resources (textbooks, printed reference cards provided by the teacher, or devices with internet access if available) to research and define the following terms: wave, wavelength (symbol λ), amplitude, period (symbol T), frequency (symbol f), wave velocity (symbol v), transverse wave, and longitudinal wave. Each learner then constructs a single A4-page labelled wave diagram in their exercise book showing BOTH a transverse wave and a longitudinal wave. On the transverse wave they label: crest, trough, amplitude, wavelength, and one full period. On the longitudinal wave they label: compression, rarefaction, and wavelength. Learners then add Kenyan-context annotations linking each term to real local examples: for wavelength they write one example for a Lake Victoria water wave (long wavelength, measured in metres) and one for a spoken Swahili vowel sound (short wavelength, measured in centimetres or millimetres). Learners compare diagrams with a partner, checking for accuracy, before a whole-class share-out in which the teacher builds a class master diagram on the board.	The teacher distributes reference cards (or directs learners to the relevant textbook pages) and says: You now have eight terms to find and understand. Do not just copy a definition. Read it, look at your slinky sketch, and then write what it means in your own words first. Then add the formal definition. You have seven minutes. [Circulates, checking progress, asking probing questions such as: How does wavelength differ from amplitude? Could a very loud sound have a short wavelength? What decides the period of a wave?] After seven minutes the teacher says: Now build your diagram. Both wave types on the same page. Label every part. Then add your Kenyan examples as annotations. You have six minutes. [WAIT TIME between prompts: 30 seconds.] During the whole-class share-out the teacher cold-calls four students to each contribute one label to the master diagram, asking: [Name], come and label one feature and tell us what it means in the context of the fisherman on Lake Victoria. The teacher closes the Explain phase by saying: Every scientific investigation needs a shared language. From today, these eight terms are OUR language for understanding how the fisherman hears thunder from behind the cliff.	Vocabulary-anchored diagram construction: linking each new term to a concrete sketch and a Kenyan real-world example simultaneously builds definitional knowledge and contextual understanding. The peer-comparison step adds a low-stakes collaborative checking routine that surfaces misconceptions (for example, confusing amplitude with wavelength) before they are reinforced.	Collect two or three learner diagrams at random (photograph with a phone if collecting is not practical) to check for: correct positioning of amplitude as measured from equilibrium not crest to trough; correct labelling of ONE full wavelength (crest to crest or equivalent); presence of Kenyan-context annotations. Provide immediate oral feedback to the whole class on the two most common errors observed, without naming individual learners. Note learners who still conflate amplitude with wavelength for targeted support in Lesson 2.
Driving	Each learner receives two sticky	The teacher says: You have two	Public question-cataloguing on the	Scan the wonder questions for

Question Board (DQB) Creation  VIDEO: Wave properties Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Landforms from Wave Erosion and Deposition (Flexbook) Source: CK-12  Search ARES for similar readings	notes (or small paper slips). On the FIRST sticky note they write one WONDER question beginning with the stem: I wonder why... or I wonder how... — the question must relate to the Lake Victoria fisherman phenomenon or to waves in their own Kenyan environment. On the SECOND sticky note they write one PREDICTION REVISION: they re-read their original prediction written in the Predict phase and write whether they want to keep it, revise it, or replace it, and briefly say why. Learners come to the front in an orderly queue and post both notes on the DQB, which is divided into four columns headed: SOUND WAVES, WATER WAVES, OTHER WAVES, and APPLICATIONS. The teacher reads three or four of the posted wonder questions aloud to the class and highlights how each question points to a future lesson in the unit (for example, a question about sound going around corners points to Lesson 5 on diffraction).	sticky notes. On the yellow one, write the question this phenomenon leaves you with. It must start with I wonder why or I wonder how. On the blue one, look back at your prediction and decide: after what you have learned today about wave vocabulary, would you change anything? Write your revision and say why. You have three minutes. [WAIT TIME: 3 minutes.] As learners post their notes the teacher reads several aloud and says: Listen to these questions. [Reads three examples.] These are the questions that will drive our investigation for the next seven lessons. We will not answer all of them today. Some you will answer in Lesson 3, some in Lesson 5, some in Lesson 7. By Lesson 8 we will come back to this board and close every question with evidence. This board belongs to you. Every time we learn something new, we add to it.	DQB creates a living record of collective curiosity that makes the unit coherent and learner-driven. By organising notes into columns linked to wave types and applications, the teacher visually maps the architecture of the entire sub-strand in a way learners can see and return to. Revision of original predictions after vocabulary instruction models the scientific practice of updating ideas in light of new evidence.	disciplinary quality: questions that identify a specific phenomenon to explain (for example, WHY does sound go around a corner but light does not?) indicate stronger inquiry readiness than purely factual questions (for example, WHAT is a wave?). Use the distribution of question types to gauge class readiness and adjust the scaffolding level in Lesson 2. The prediction-revision notes reveal whether learners have integrated the new vocabulary — a revision that simply adds words like amplitude without changing the conceptual content suggests surface-level uptake.
Model Building Phase  VIDEO: Wave properties Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Landforms from Wave Erosion and Deposition (Flexbook) Source: CK-12  Search ARES for similar	In the final three minutes of the lesson, learners return to their labelled wave diagram and add a small sketch in one corner: they draw the Lake Victoria fisherman scenario as a simple scene (shoreline, cliff or headland, fisherman, storm in the distance) and draw an arrow or curved line to show the path of the thunder sound reaching the fisherman. They label this path with one question mark, indicating that they do not yet know the scientific explanation for why the sound curves around the cliff. This partial model will be added to and refined in every subsequent lesson,	The teacher says: Before we finish, add a small scene sketch to your diagram: the fisherman, the cliff, the storm. Draw the path you think the sound is taking. Do NOT worry about being wrong. This is your starting model. You will improve it every lesson. Write at the bottom: This is my model on [date] and it is not complete yet. [WAIT TIME: 3 minutes.] The teacher closes the lesson by saying: Today you have met the phenomenon that will drive our whole unit. You have the vocabulary. You have your first wonder questions on the board. Over the next seven lessons you	Partial model construction at the START of the unit — rather than at the end — positions learners as scientists who refine models iteratively rather than students who produce a correct answer once. The explicit labelling of the model as incomplete and dated creates a growth mindset around model revision and gives each learner a personalised artefact that connects every subsequent lesson to the anchoring phenomenon.	A quick exit-check: as learners leave (or in the final 60 seconds), the teacher asks three randomly selected learners to state ONE wave vocabulary term and give a Kenyan example of it without looking at their notes. This verbal exit ticket takes under 90 seconds and gives the teacher a spot-check on retention of the lesson core. Record which terms are recalled fluently (likely amplitude and wavelength) and which are not (likely period and frequency) to inform the pacing of Lesson 2.

readings	becoming a complete annotated explanation by Lesson 8. Learners write at the bottom of the diagram: This is my model on [date]. It is NOT complete yet.	are going to build the science that explains every puzzling thing that fisherman noticed — and you are going to connect it to mobile phones in the Rift Valley, ultrasound at Kenyatta National Hospital, and protecting our coastline at Mombasa and Lamu. Hold onto your questions. The best scientists are the ones who keep asking.		
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D. TEACHER REFLECTION

After the lesson, consider the following questions: Did every learner produce a labelled diagram with at least five correct features? Were the wonder questions on the DQB genuinely open and phenomenon-linked, or were they closed factual questions that suggest learners did not engage with the puzzling scenario? Did the cold-calling distribute across gender, seating position, and apparent confidence level? Were there learners who could not yet distinguish amplitude from wavelength — and what additional support will they receive at the start of Lesson 2? Did the Kenyan-context annotations (Lake Victoria waves, Swahili vowel sounds) feel meaningful to learners or superficial? If superficial, consider inviting learners to suggest their own local examples before imposing the teacher examples in future iterations. Finally, review the DQB: does the distribution of wonder questions across the four columns indicate a balanced class curiosity, or are all questions clustered in one area (for example, all about sound, with no questions about water or other waves)? Use this distribution to deliberately build in cross-wave comparisons during Lessons 2 and 3.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	A narrated scenario of a Kisumu fisherman hearing thunder from behind a cliff and seeing water waves change direction near the shore, illustrated by a photograph of the Kisumu shoreline and a slinky spring demonstration of transverse and longitudinal pulses
What did I learn?	The key vocabulary of waves: wavelength, amplitude, period, frequency, wave velocity, transverse wave, and longitudinal wave, each anchored to a labelled diagram and annotated with Kenyan real-world examples including Lake Victoria water waves and spoken Swahili vowel sounds
How does this explain the phenomenon?	Learners produced initial (pre-instruction) predictions for the anchoring phenomenon and acknowledged that the new vocabulary gives them a shared scientific language to begin investigating the fisherman puzzle, while recognising that the full explanation (reflection, refraction, and diffraction) will emerge across the unit

LESSON 2 (80 minutes): The Wave Equation — Connecting Speed, Frequency and Wavelength

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners derive and apply the wave equation $v = f\lambda$ using Kenyan contexts, building the quantitative foundation needed to explain why the fisherman's sound bends around the headland, why water waves change direction at the shallow bay, and why a mobile signal fades behind a Rift Valley hill.
Knowledge	Learners explain the terms frequency (f), period (T), wavelength (λ) and wave speed (v); state the relationship $T = 1/f$; interpret and apply the wave equation $v = f\lambda$ to calculate unknown wave characteristics.
Skills	Learners measure period and frequency from a slinky/ripple-tank demonstration; derive $v = f\lambda$ through guided questioning; solve four Kenya-contextualised numerical problems (Nairobi FM radio wavelength, Mombasa echo speed, Lake Victoria boat-engine vibration frequency, Indian Ocean coastal swell period).
Attitudes	Learners appreciate that mathematical relationships in physics are tools for explaining real-world phenomena, and that the same equation governs sound, water, and radio waves in Kenyan environments.
Key Inquiry Question	How are waves applicable in our day-to-day life?
Purpose in Storyline	Lesson 1 introduced wave vocabulary and the anchoring phenomenon. Lesson 2 gives learners the quantitative language — the wave equation — so they can begin to assign numbers to the fisherman's observations: How fast is the thunder sound travelling? What is the wavelength of the radio signal that fades behind a hill? This numerical precision will be the scaffold for Lessons 3–8, where reflection, refraction, and diffraction are explored in depth.
Safety Notes	Handle slinky springs carefully — do not overstretch or release suddenly, as the ends can recoil and cause eye injury. Keep the slinky flat on the floor away from walkways. If a ripple tank is used, wipe up any water spills immediately to prevent slipping. Learners with photosensitive epilepsy should be warned before stroboscopic ripple-tank lighting is used.

B. LESSON OVERVIEW

In this 80-minute lesson learners move from qualitative wave vocabulary (Lesson 1) to quantitative understanding. A teacher-led slinky demonstration on the floor and a ripple-tank or simulation observation provide concrete evidence that a higher-frequency source produces shorter-wavelength waves at the same medium speed. Through structured guided questioning with mandatory 30-second wait times and a minimum of four cold-call responses per key step, learners co-derive the wave equation $v = f\lambda$ and its rearrangements. They then apply the equation to four Kenya-contextualised numerical problems before updating the class Driving Question Board and revising their anchoring-phenomenon predictions.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
1 — PREDICT (15 minutes)	Learners individually read a two-sentence scenario card (distributed by the teacher): 'A	Distribute scenario cards face-down; say: 'Do NOT flip the card until I say go.' After settling, say:	Prediction-before-observation (Elicit Prior Knowledge): surfacing and publicly recording divergent	Circulate during individual writing. Look for: (1) correct use of at least two Lesson-1 vocabulary terms;

 VIDEO: Wave properties Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Landforms from Wave Erosion and Deposition (Flexbook) Source: CK-12  Search ARES for similar readings	fisherman on Lake Victoria hears a boat engine vibrating steadily. He counts roughly 4 complete wave crests hitting his canoe every second. The crests are about 0.5 m apart. His friend, standing further along the shore, says the waves where he is standing seem to be the same speed but closer together.' Learners write individual predictions in their science notebooks: (a) What is the speed of these water waves? (b) If the friend observes crests only 0.25 m apart, what must have changed — frequency, speed, or wavelength? (c) Write your prediction as a sentence using at least two wave vocabulary words from Lesson 1. Learners share predictions with a partner (Think-Pair) before a brief whole-class share.	'Flip and read silently — you have 2 minutes.' [WAIT 2 minutes.] Then: 'Write your three predictions now — do not discuss yet.' [WAIT 4 minutes.] Prompt the Think-Pair: 'Turn to your neighbour and compare predictions — you have 90 seconds.' Cold-call 3 different pairs to share one prediction each. Record contrasting predictions (e.g. 'speed changes' vs 'frequency changes') on the board under the heading BEFORE WE LOOK AT EVIDENCE. Do NOT correct — say: 'We are going to test these ideas right now.'	ideas creates cognitive dissonance that motivates close observation. Recording predictions on the board also creates accountability — learners know they will return to these claims.	(2) whether learners predict that speed stays constant while frequency and wavelength vary inversely. Flag learners who write 'speed increases with frequency' — these learners need targeted attention during the derivation phase. Note names for cold-calling during Explain.
2 — OBSERVE (20 minutes)  VIDEO: Wave properties Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Landforms from Wave Erosion and Deposition (Flexbook) Source: CK-12  Search ARES for similar readings	PART A — Slinky Demonstration (10 min): The teacher stretches a slinky along the floor of the classroom (approximately 4 m). Two learner volunteers hold each end. The teacher generates slow transverse pulses (approximately 1 Hz), then faster pulses (approximately 3 Hz). The rest of the class observes and records in a two-column table: 'What I notice about the distance between crests (wavelength)' and 'What I notice about how often crests pass a fixed point (frequency)'. PART B — Ripple Tank / Simulation (10 min): The teacher projects a ripple-tank simulation (e.g., PhET Wave Interference, or a pre-downloaded offline Kenya Education Cloud video of a ripple tank). The simulation shows: (i) low-frequency source → widely spaced wavefronts; (ii) high-frequency source → closely spaced	SLINKY: Recruit two volunteers — say: 'Amina and Kamau, please hold the ends and keep tension constant — do not walk closer or further.' Generate slow pulses and ask the class: 'Count how many full waves fit along the spring — whisper the number to yourselves.' [WAIT 10 seconds.] Cold-call 2 learners. Then increase frequency and ask: 'What happened to the number of waves along the spring?' [WAIT 15 seconds.] Cold-call 2 learners. Ask: 'Did either of you move? Did the speed of my hand change the speed of the wave along the spring, or only how often I sent pulses?' [WAIT 30 seconds.] Cold-call 2 more learners. SIMULATION: As the simulation runs, narrate: 'Watch the wavefronts — I am only changing how fast the source vibrates.' Pause at each frequency and ask: 'Tell your partner: what	Observation-with-structured-recording: the two-column table forces learners to separate what they observe (wavelength, frequency) from what they infer (speed). This disciplined separation is the NGSS Science and Engineering Practice SEP-3 (Planning and Carrying Out Investigations) at the observation level, preparing learners to construct an explanation from evidence.	Collect notebooks briefly or do a quick show-hands: 'Thumb up if your table shows that wavelength decreased when frequency increased.' Address any thumbs-down immediately. Check that all learners have labelled λ correctly on their sketches — a common error is labelling only half a wavelength (crest to trough).

	wavefronts; (iii) the wave speed (indicated on screen or calculated from the simulation scale) remains constant in the same medium. Learners sketch both patterns in their notebooks, labelling λ (wavelength) and the direction of wave travel. They note the wave speed value in each case.	changed and what stayed the same?' [WAIT 20 seconds.] Cold-call 3 pairs. Record agreed observations on the board: 'Higher frequency $\rightarrow$ shorter wavelength; speed appears constant in same medium.'		
3 — EXPLAIN (30 minutes)  VIDEO: Wave properties Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Landforms from Wave Erosion and Deposition (Flexbook) Source: CK-12  Search ARES for similar readings	DERIVATION (10 min): Guided by the teacher's questioning sequence, learners co-derive $v = f\lambda$ in their notebooks step by step. They write each logical step as prompted, with 30-second wait times between steps. NUMERICAL PROBLEMS (20 min): Learners work individually (then compare with a neighbour) on four Kenya-contextualised problems printed on a worksheet: PROBLEM 1 — Radio waves (Nairobi FM): 'Classic 105 FM Nairobi broadcasts at 98.4 MHz. Radio waves travel at 3.0×10^8 m/s. Calculate the wavelength of the broadcast signal. What does this wavelength tell us about whether the signal can diffract around a hill in the Rift Valley?' PROBLEM 2 — Echo in Mombasa: 'A tourist at Fort Jesus in Mombasa claps once and hears an echo from the opposite wall 0.12 s later. The wall is 20.4 m away. Calculate the speed of sound. Compare your answer to 340 m/s — what might explain any difference?' PROBLEM 3 — Boat engine on Lake Victoria: 'A fisherman near Kisumu measures the wavelength of water waves produced by his boat engine as 0.8 m. The waves travel at 3.2 m/s. Calculate the frequency of the engine vibration and its period.' PROBLEM 4 — Indian Ocean	DERIVATION sequence (each step: pose question $\rightarrow$ WAIT 30 seconds $\rightarrow$ cold-call minimum 4 students): Step 1: 'In one full period T, how far does the wave travel?' [Expected: one wavelength, λ.] Step 2: 'Write that as an equation using distance = speed $\times$ time.' [Expected: $\lambda = v \times T$.] Step 3: 'We know $T = 1/f$. Substitute that in.' [Expected: $\lambda = v/f$.] Step 4: 'Rearrange to make v the subject.' [Expected: $v = f\lambda$.] Write each step on the board only AFTER 4 cold-calls. Say: 'We have just derived this — you did not memorise it, you built it.' PROBLEMS: Say: 'Problem 1 is about a radio station you may have heard on the matatu to school. Work alone for 8 minutes — show every step.' [Circulate; look for unit errors, especially MHz $\rightarrow$ Hz conversion.] After 8 minutes: 'Compare with your neighbour — do your answers agree?' Cold-call one pair per problem. For Problem 2 ask: 'Why divide by 2? Cold-call.' [Expected: sound travels to wall AND back.] For Problem 4 ask: 'Why does the period of an Indian Ocean swell matter to a fisherman launching a boat?' [WAIT 30 seconds.] Cold-call 3 learners — connect back to the anchoring phenomenon.	Guided Derivation followed by Contextualised Problem-Solving: deriving the equation from first principles (SEP-6: Constructing Explanations) gives learners ownership of the mathematics. The four Kenya problems apply the equation in contexts that span the anchoring phenomenon's domains (water, sound, radio), helping learners see that one equation governs multiple observations. This cross-context transfer is the core of disciplinary sensemaking in physics.	Collect one problem (Problem 3 — Lake Victoria) at the end of this phase as an exit slip. Mark criterion: (1) correct formula stated; (2) correct substitution with units; (3) correct numerical answer for f (4 Hz) and T (0.25 s); (4) units carried through. This slip informs grouping and targeted support for the next lesson. Also monitor: do learners correctly convert MHz to Hz? This is the highest-frequency error observed in Kenyan Form 3/4 radio-wave problems.

	swell: 'A coastal engineer at Mombasa Marine Reserve measures ocean swells with a wave speed of 15 m/s and a wavelength of 75 m. Calculate the period and frequency of the swell. How many crests hit the shore per minute?' Learners check answers with a neighbour, then the teacher reveals model solutions step by step on the board.			
4 — DRIVING QUESTION BOARD UPDATE (8 minutes)  VIDEO: Wave properties Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Landforms from Wave Erosion and Deposition (Flexbook) Source: CK-12  Search ARES for similar readings	Learners receive two sticky notes each (one green, one orange). On the GREEN note they write one thing the wave equation NOW explains about the anchoring phenomenon (e.g., 'The thunder sound has a long wavelength because its frequency is low — this may help it bend around the headland'). On the ORANGE note they write one NEW question the wave equation raises for them (e.g., 'If the speed of sound changes in different temperatures, does $v = f\lambda$ mean the wavelength changes too?'). Learners post both notes on the class Driving Question Board in the designated 'Lesson 2 — Wave Equation' column. The teacher reads three green and three orange notes aloud, paraphrasing each for the class.	Say: 'Look at the anchoring phenomenon written at the top of our Driving Question Board — the fisherman, the thunder, the bending water waves, the fading Rift Valley signal. Write on your GREEN note: what does $v = f\lambda$ now let you explain or calculate about that scene?' [WAIT 3 minutes for writing.] As learners post notes, scan for the most insight-rich and most confused responses. Read three green notes aloud: 'Here is what Akinyi wrote...' Paraphrase, then say: 'Does everyone agree this follows from $v = f\lambda$?' [WAIT 15 seconds.] Read three orange notes: 'Great question from Otieno — if speed changes between deep and shallow water, does wavelength change while frequency stays the same? We will investigate that in Lesson 4.' Update the board header: 'Lesson 2 ANSWERED: How are f , λ , and v related mathematically?'	Driving Question Board (DQB) — Public Knowledge-Building: posting green (explained) and orange (new questions) notes externalises each learner's current model of the phenomenon, makes collective progress visible, and seeds the inquiry questions for future lessons. This is aligned with NGSS SEP-7 (Engaging in Argument from Evidence) at the communication level.	Observe the green notes: do learners correctly connect the wave equation to the anchoring phenomenon, or do they write generic statements (' $v = f\lambda$ is useful')? Generic responses indicate surface-level engagement; phenomenon-connected responses indicate deeper sensemaking. Photograph the board for the teacher's lesson reflection record.
5 — MODEL BUILDING (7 minutes)  VIDEO: Wave properties Source: Khan Academy (English - US curriculum)  Search ARES	Learners open their anchoring-phenomenon model drawing from Lesson 1 (a sketch of the Lake Victoria scene). They add a new layer using a different colour pen or pencil: (a) Label the thunder sound waves with an estimated long wavelength (e.g., ' $\lambda \approx 34$ m if f	Say: 'Take out your Lesson 1 phenomenon model. Use a different colour — add numbers. Where are the sound waves? Give them a wavelength using $v = f\lambda$. Where are the water waves from the storm? Give them a wavelength.' [WAIT 5 minutes.]	Cumulative Model Revision with Intentional Incompleteness: annotating the model with calculated values embeds the wave equation into the phenomenon, making mathematics visually purposeful. The deliberately incomplete	Check that learners have (a) added at least one wavelength annotation with a calculation shown, (b) used $v = f\lambda$ correctly in the annotation, and (c) articulated a specific gap in their understanding in the incomplete sentence. Learners who write

for similar videos HTML: Landforms from Wave Erosion and Deposition (Flexbook) Source: CK-12  Search ARES for similar readings	= 10 Hz, $v = 340 \text{ m/s}$) and the boat-engine water waves with a shorter wavelength (e.g., $\lambda = 0.8 \text{ m}$); (b) Add a small ' $v = f\lambda$ ' annotation box in the corner of the diagram; (c) Write one sentence beneath the diagram that begins: 'I can now calculate _____ but I still cannot explain why the waves bend around the headland because _____. ' The incomplete sentence is intentional — it anchors curiosity for Lessons 3 and 4.	Circulate and prompt with: 'What frequency would a deep rumble of thunder have — very high or very low? So what does that tell you about its wavelength?' [WAIT 15 seconds per learner before moving on.] At the 5-minute mark: 'Everyone finish the sentence at the bottom — it must have a blank in it.' Cold-call 2 learners to read their sentence aloud. Say: 'Hold on to that blank. Lessons 3 and 4 will fill it.'	sentence is a cognitive hook — it prevents premature closure and sustains inquiry motivation across the lesson sequence. This reflects NGSS SEP-2 (Developing and Using Models).	vague gaps ('I don't understand waves') need one-to-one prompting to identify the specific mechanism they are missing (wave bending/diffraction).
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record brief notes in your journal: (1) During the slinky demonstration, did learners clearly observe the inverse relationship between frequency and wavelength, or was the spring motion ambiguous? If unclear, plan to use the ripple-tank simulation more prominently in the next iteration. (2) Which of the four numerical problems caused the most difficulty? The MHz-to-Hz conversion in Problem 1 is historically the most common error in Kenyan classrooms — if more than one-third of learners got it wrong on the exit slip, open Lesson 3 with a 5-minute unit-conversion warm-up. (3) Were the 30-second wait times observed consistently? Note the names of learners cold-called and ensure different learners are called in Lesson 3. Aim for full class coverage across two lessons. (4) Did the DQB orange notes reveal genuine curiosity about wave bending, or did learners merely restate the lesson content? If orange notes were superficial, the phenomena connection needs strengthening — consider showing a 90-second video clip of ocean waves refracting at the Mombasa coast at the start of Lesson 3. (5) Review the Problem 3 exit slips before Lesson 3: group learners who could not correctly derive f and T together for a peer-support activity at the start of the next lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	During the slinky demonstration on the classroom floor, learners observed that increasing the rate of hand oscillation (frequency) produced more crests along the spring length — the distance between crests (wavelength) became shorter. The overall wave appeared to travel along the spring at the same speed regardless of how fast the hand moved. In the ripple-tank simulation, learners observed that low-frequency sources produced widely spaced wavefronts while high-frequency sources produced closely spaced wavefronts, with the wave speed remaining constant in the same medium.
What did I learn?	Through guided derivation, learners established that in one period T , a wave travels exactly one wavelength λ , giving $\lambda = vT$. Substituting $T = 1/f$ yields the wave equation $v = f\lambda$. This means that in a given medium (constant v), frequency and wavelength are inversely proportional. Learners applied this to four Kenyan contexts: the wavelength of Classic 105 FM (98.4 MHz) radio waves is approximately 3.05 m; the speed of sound at Fort Jesus Mombasa is approximately 340 m/s; a Lake Victoria boat-engine wave ($\lambda = 0.8 \text{ m}$, $v = 3.2 \text{ m/s}$) has frequency 4 Hz and period 0.25 s; and Indian Ocean swells ($v = 15 \text{ m/s}$, $\lambda = 75 \text{ m}$) have period 5 s, frequency 0.2 Hz, meaning 12 crests per minute reach the shore.
How does this explain the phenomenon?	The wave equation $v = f\lambda$ explains why the fisherman at Lake Victoria can begin to assign numbers to his observations: the deep-rumble thunder sound has a low frequency and therefore a long wavelength, which — as learners will discover in Lesson 3 — enables it to diffract (bend) around the rocky headland more effectively than high-frequency sounds. The same equation explains why a Nairobi FM radio signal has a wavelength of roughly 3 m, which is relevant to whether that signal can diffract around a Rift

	Valley hill (a question that will be quantitatively resolved in Lesson 5 on diffraction). Learners cannot yet fully explain the bending mechanism, but they now have the mathematical tool to characterise any wave in the anchoring phenomenon.
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LESSON 3 (80 minutes (2 × 40-minute lessons)): Reflection of Waves: Why Does Sound Bounce Back — and Could It Explain the Fisherman's Puzzle?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To demonstrate, measure, and explain the reflection of waves, and to connect the law of reflection to real-world phenomena including the fisherman's experience on the shores of Lake Victoria.
Knowledge	Learners explain the law of reflection (angle of incidence = angle of reflected angle), describe echo formation, and calculate the speed of sound using echo timing data collected from the school perimeter wall.
Skills	Learners carry out an echo experiment (measuring distances, timing echoes, calculating sound speed); draw accurate incident and reflected ray diagrams using a protractor; and record and compare group data in a results table.
Attitudes	Learners appreciate how the scientific principle of reflection underpins everyday technologies (sonar, radar, ultrasound in hospitals) and natural phenomena experienced by Kenyan communities, demonstrating curiosity and collaborative responsibility during the experiment.
Key Inquiry Question	How are waves applicable in our day-to-day life? How do waves affect our environment?
Purpose in Storyline	Lesson 3 takes the first major step toward resolving the fisherman's puzzle: learners gather real acoustic evidence that sound bounces off hard surfaces (the school wall as a proxy for the rocky headland), measure the effect, and formalise it as the law of reflection. By the end of the lesson learners can explain part of the puzzle — reflected sound can reach the fisherman from behind the cliff — but the DQB update makes clear that reflection alone cannot explain sound bending around the headland, setting up the need for diffraction in Lesson 4.
Safety Notes	Learners must not stand closer than 15 m to the perimeter wall during the echo experiment to allow the echo to be distinguishable from the original clap. Groups must nominate a safety monitor to check the area behind the wall is clear before clapping. No learner should cup hands to ears and clap simultaneously — designate separate clappers and listeners. During ray-diagram work, protractors have sharp edges; handle with care.

B. LESSON OVERVIEW

Learners begin by predicting what happens to a sound pulse and a torch-beam when they strike a flat versus a curved wall — connecting to the fisherman's cliff. Groups then move to the school perimeter wall to carry out the echo experiment: standing at measured distances (20 m, 30 m, 40 m), clapping rhythmically and timing echoes with phone stopwatches or counting beats to calculate the speed of sound. Results from all groups are pooled and compared. Back in the classroom, learners draw incident and reflected ray diagrams for sound bouncing off Nairobi building walls and the Lake Victoria headland, and the teacher formalises the law of reflection using protractor measurements from the diagrams. The lesson closes with a Driving Question Board update that celebrates the new explanatory power of reflection while honestly flagging what it cannot yet explain — the sound going around the headland — sustaining the storyline's productive tension.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Phase 1 — Predict (15 minutes)  VIDEO: Central limit theorem Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Reflection Symmetry (Flexbook) Source: CK-12  Search ARES for similar readings	Learners sit in their groups (4–5 per group). Each learner first writes an individual silent prediction in their science journal responding to two prompts displayed on the board: (1) 'A friend stands 30 m from the school's concrete perimeter wall and claps once. Draw or describe what you think happens to that sound pulse when it hits the wall.' (2) 'A torch-beam shines at a flat wall at an angle — draw the path of the light before and after it hits the wall. Now imagine the wall is curved like the headland at Lake Victoria — what changes?' After 4 minutes of silent writing, learners share predictions with their group partner, then three groups are cold-called to share with the class. Learners mark their own prediction sheets with a star on the idea they are most confident about and a question mark on what they are least sure of.	Display the two prompts. Say: 'Before we go outside, I want each of you — individually, silently — to write or draw your own prediction. No conferring yet. You have 4 minutes. Go.' [Start timer. Circulate, reading over shoulders. Do NOT correct — note 2–3 interesting or conflicting predictions to return to later.] After 4 minutes: 'Now share your prediction with the person next to you. You have 90 seconds.' [Allow pair talk.] Then cold-call exactly 3 groups: 'Group 2 — Amina, tell us your group's best prediction for what happens to the sound at the wall.' [WAIT 12 seconds.] 'Group 4 — Kipchoge, does your prediction agree or disagree with Group 2? Why?' [WAIT 12 seconds.] 'Group 6 — Zawadi, what did you draw for the curved wall?' [WAIT 10 seconds.] Record key prediction phrases verbatim on the board in a two-column T-chart labelled FLAT WALL / CURVED WALL. Do not evaluate — say: 'We will come back to these after the experiment and see whose prediction the evidence supports.'	Individual prediction journaling followed by pair-share and class T-chart. This activates prior knowledge of bouncing (balls, light in mirrors) and creates a public record of class thinking that learners will revise with evidence — a core element of the NGSS Practice of asking questions and defining problems, and of the ARES predict–observe–explain cycle.	Teacher scans journals during silent writing: look for whether learners spontaneously use the word 'bounce' or draw an angled arrow — this reveals prior knowledge of reflection. Flag learners who draw the sound 'stopping' at the wall or 'going through' it — these misconceptions will be directly addressed in the Explain phase. Observable indicator: every learner has a drawn or written prediction on paper before leaving for the outdoor experiment.
Phase 2 — Observe: The Echo Experiment (25 minutes)  VIDEO: Central limit theorem Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Reflection Symmetry	Groups move to the school perimeter wall (a hard concrete or brick surface, at least 40 m of clear space in front of it). Each group is assigned one distance: Group A = 20 m, Group B = 25 m, Group C = 30 m, Group D = 35 m, Group E = 40 m (adjust for available groups). The distance is measured with a tape measure or measured rope and marked with chalk or a stone. Within each group: one learner is the Clapper, one is the Timer (phone stopwatch or counting rhythmic beats), two are Listeners, and one is the Recorder.	Before going outside: distribute the Results Recording Sheet (prepared in advance or quickly drawn in journals) and read the safety briefing aloud: 'Safety monitor for each group: check there is nobody behind the wall before we start. Clappers and listeners are different people. Minimum distance from the wall is 15 m — Group A, you are at 20 m, that is your minimum. Any questions?' [WAIT 10 seconds.] Outside: circulate between groups during the experiment. When visiting Group A say: 'Tell me what	Quantitative field investigation with multi-group data pooling (the NGSS Practice of Planning and Carrying Out Investigations, and Analysing and Interpreting Data). By pooling data from 5 distances, learners see that the same physical constant emerges despite variation — building understanding that reflection of sound is a repeatable, measurable phenomenon. The directional pointing exercise ('where did the sound come from?') directly primes the concept of incident and reflected angles.	Each group's completed results table is the observable product. Teacher checks: (a) Is the formula $2d \div (t/10)$ applied correctly? (b) Is the calculated speed in a plausible range (300–380 m/s)? Groups outside this range are flagged for a re-check of their timing method during the Explain phase. Listening indicator: when asked 'what direction does the echo come from?' — learners who point toward the wall have understood directional reflection; learners who point upward or are confused will need the ray diagram scaffolding.

(Flexbook) Source: CK-12 Search ARES for similar readings	Procedure: (i) The Clapper claps a sharp single clap; Listeners confirm they hear the echo; (ii) The Clapper then claps rhythmically — adjusting the clapping rate until each new clap coincides exactly with the echo of the previous one; (iii) The Timer measures the time for 10 claps (= 10 echo round-trips) and records it; (iv) Each group repeats 3 times and averages. The Recorder fills in the group results table: Distance (d), Time for 10 echoes (t), Time for 1 echo round-trip (t/10), Speed of sound = $2d \div (t/10)$. Groups rotate roles once. All groups return to class after 20 minutes and post their calculated speed of sound on the class results board (a large sheet of manila paper or the chalkboard).	you are hearing — describe the sound that comes back.' [WAIT 12 seconds.] When visiting Group C say: 'What direction does the echo seem to come from? Point to where you think the sound came from.' [WAIT 10 seconds.] This primes directional thinking for the ray diagram. If a group cannot detect an echo (common in windy conditions or near vegetation), prompt: 'Move 5 m closer. What changes?' Back in the classroom, point to the class results board: 'Groups, look at all the calculated speeds. What do you notice? Are they similar or very different? Why might there be variation?' Cold-call 2 groups for answers. [WAIT 12 seconds each.] Then reveal: 'The accepted speed of sound in air at room temperature is approximately 340 m/s. How close are your values? What sources of error could explain differences?' Record learner-generated sources of error on the board.		
Phase 3 — Explain: The Law of Reflection and the Fisherman's Puzzle (20 minutes)  VIDEO: Central limit theorem Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Reflection Symmetry (Flexbook)	Back in class, learners open their journals to a fresh page and draw two diagrams side by side. Diagram 1: A sound pulse from a person standing in front of a flat Nairobi building wall — draw the incident ray, the normal (dashed perpendicular line), and the reflected ray. Label the angle of incidence (θ_i) and angle of reflection (θ_r). Use a protractor to measure both angles. Diagram 2: The fisherman on the shore of Lake Victoria, with the rocky headland as a curved surface. Draw two incident sound rays from the storm hitting different parts of the headland, draw their reflected rays, and use arrows to show where the reflected sound travels	Say: 'We now have evidence that sound bounces off the school wall. Let's use that evidence to draw the picture scientifically.' Display a partially drawn ray diagram on the board — incident ray hitting a flat surface — and ask: 'What is the normal? Someone come and draw it.' [WAIT 12 seconds. Accept a volunteer.] Then: 'Now measure the angle of incidence with your protractor. Tell me — if the angle of incidence is 40°, what do you predict the angle of reflection will be?' [WAIT 10 seconds. Cold-call 2 learners before confirming.] After learners draw Diagram 2, ask: 'Kipchoge — in your diagram, does any reflected ray from the headland reach the fisherman	Diagrammatic modelling with protractor measurement (NGSS Practice: Developing and Using Models) combined with anchoring-phenomenon connection. By drawing the fisherman scenario explicitly, learners apply the abstract law to the original puzzle, making the explanation personally meaningful. The 'annotate your original prediction' activity is a metacognitive strategy (Learning to Learn competency) that makes conceptual change visible to learners themselves.	Teacher circulates and checks: (a) Is the normal drawn perpendicular to the surface — not vertical? Many learners default to drawing the normal as a vertical line regardless of surface orientation. (b) Do measured angles in Diagram 1 satisfy $\theta_i = \theta_r$ within $\pm 3^\circ$? (c) In Diagram 2, do reflected rays point in physically plausible directions (not back into the headland)? Observable indicator: every learner has a completed 2-sentence journal explanation of what reflection can and cannot explain about the fisherman's puzzle before the DQB phase begins.

Source: CK-12 Search ARES for similar readings	— does any reflected ray head toward the fisherman's position? Learners compare diagrams in pairs, then the class discusses: 'Does reflection alone explain how the fisherman hears the thunder?' Learners write a 2-sentence explanation in their journals. The teacher then formalises the law of reflection verbally and on the board: 'The angle of incidence equals the angle of reflection — both measured from the normal to the surface at the point of contact.' Learners measure angles from their own Diagram 1 and verify the law. Finally, learners revisit their Phase 1 predictions and annotate them: circle what was correct, cross out what was wrong, and add the scientific term.	directly?' [WAIT 12 seconds.] Then: 'Amina — could reflected sound explain ALL of the fisherman's puzzle — including the sound coming from behind the cliff where there is no straight-line path and no reflective surface he can see?' [WAIT 15 seconds.] Guide the class to the honest answer: reflection explains echoes and some redirection, but cannot explain sound reaching a completely shadowed zone behind the headland. Say: 'Write this in your journal: reflection is part of the answer — but it is not the whole answer. What other wave behaviour might we need?' This is the productive tension that will drive Lesson 4.		
Phase 4 — Driving Question Board Update (10 minutes) VIDEO: Central limit theorem Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Reflection Symmetry (Flexbook) Source: CK-12 Search ARES for similar readings	Each learner writes two sticky notes (or two entries in a designated section of their journal if sticky notes are unavailable): (1) A CLAIM card — one sentence stating what they now know about the fisherman's puzzle because of today's lesson (e.g. 'Reflected sound from the cliff face could send some of the thunder's sound toward the fisherman even if the storm is partly behind the cliff.'). (2) A NEW QUESTION card — one question that today's evidence raises but does not answer (e.g. 'But what about the sound reaching the completely shadowed cove — no wall is facing that way?'). The class DQB (a large sheet of paper or section of the chalkboard divided into columns: WHAT WE OBSERVED / WHAT WE THINK WE KNOW / WHAT WE STILL WONDER) is updated. The teacher reads 3 CLAIM cards and 3 NEW QUESTION cards	Say: 'Take 3 minutes — silently write your CLAIM card and your NEW QUESTION card. No talking.' [Circulate. After 3 minutes:] 'Post your cards on the DQB now.' [Give 1 minute.] Read 3 claim cards aloud, pausing after each: 'Thumbs up if you agree with this claim, sideways if you are not sure, down if you disagree. [WAIT 8 seconds for thumbs.] Why did some of you put thumbs sideways — Zawadi?' [WAIT 12 seconds.] After reading 3 new question cards, circle the ones related to sound going around the headland and say: 'Look at this question — it is exactly what we cannot explain yet with reflection. Keep it in mind. Lesson 4 is going to attack this question directly.' Write on the DQB in a distinct colour: 'Reflection explains part of the fisherman's puzzle — but NOT how sound goes AROUND the headland.'	Claim–Evidence–Reasoning (CER) sticky-note protocol on the Driving Question Board. This strategy makes learners articulate the boundary of their current understanding — distinguishing what reflection explains from what it does not — which is epistemically honest science practice (NGSS Practice: Engaging in Argument from Evidence). The public voting ritual builds a shared class understanding and social accountability for the evolving model.	Teacher reads all posted CLAIM cards after class. Look for: (a) claims that conflate reflection with the complete explanation of the fisherman's puzzle — these learners need targeted questioning in Lesson 4; (b) NEW QUESTION cards that already anticipate diffraction or bending ('What if the wave goes around the edge?') — these learners can be called on as 'thinking partners' in Lesson 4's predict phase. Observable indicator: every learner has at least one CLAIM and one NEW QUESTION posted on the DQB.

	aloud, and the class votes (thumbs up/sideways/down) on whether they agree with each claim. New questions that the class agrees are not yet answered are circled in a different colour — these become the explicit bridge to Lesson 4 (diffraction).			
Phase 5 — Model Building (10 minutes)  VIDEO: Central limit theorem Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Reflection Symmetry (Flexbook) Source: CK-12  Search ARES for similar readings	Learners return to the cumulative wave model they began building in Lesson 1 (a large annotated diagram of the Lake Victoria scene: fisherman, storm, headland, bay, shallow water, deep water). Today they add a new layer using a different colour pencil or pen: (i) Draw at least 2 incident sound ray arrows from the storm toward the rocky headland, (ii) Draw the corresponding reflected ray arrows using the correct angle of reflection, (iii) Add a label: 'REFLECTION — sound bounces off the headland; some reaches the fisherman, but reflected rays do NOT reach into the shadowed cove.' (iv) In a model legend/key box in the corner, write: 'Reflection: angle of incidence = angle of reflection.' Learners compare their updated model with a neighbour and check: do the reflected rays point in the right direction? Each learner writes one sentence at the bottom of their model page: 'My model can now explain _____ but still cannot explain _____.'	Say: 'Get out your cumulative Lake Victoria model — the diagram we started in Lesson 1. Use a new colour for today's addition so we can see how our model is growing lesson by lesson.' [Allow 7 minutes for individual model update.] After 5 minutes, say: 'Swap with your neighbour. Check one thing: do their reflected rays obey the law of reflection — does the angle of incidence look equal to the angle of reflection? If not, point it out kindly.' [Allow 2 minutes of peer review.] Then: 'Read me one sentence from the bottom of your model — what can your model explain, and what can it still not explain? Achieng — go.' [WAIT 12 seconds.] 'Mwangi — same question, different answer if possible.' [WAIT 12 seconds.] Close with: 'Every lesson, this model grows. By Lesson 8, it will explain the whole fisherman's puzzle — and the mobile-phone signal in the Rift Valley. Today we added reflection. Next lesson: we tackle the mystery the DQB flagged — how waves go around corners.'	Cumulative layered model revision (NGSS Practice: Developing and Using Models). By using a new colour for each lesson's additions, learners physically see their scientific model growing in explanatory power — a metacognitive record of learning that directly supports the 'Learning to Learn' core competency. The sentence frame 'can explain / still cannot explain' enforces epistemic precision about the scope of each concept.	Collect or photograph learner models at the end of the lesson. Check: (a) Are reflected rays drawn at the correct angle (roughly equal to incident angle)? (b) Does the annotation explicitly acknowledge that the shadowed cove is NOT explained by reflection? (c) Is the new layer in a distinct colour, showing the learner is tracking cumulative model growth? Learners whose models show reflected rays entering the shadowed cove have a persistent misconception about reflection that must be corrected before Lesson 4 introduces diffraction.

D. TEACHER REFLECTION

After the lesson, consider the following questions: (1) During the echo experiment, did all groups obtain a speed of sound within a reasonable range (300–380 m/s)? If not, what was the most common source of error — clapping rhythm, timing over too few echoes, or background noise — and how will you address this in a follow-up? (2) In the Explain phase, how many learners still believed after the lesson that reflection alone could explain sound reaching the completely shadowed cove? If more than

30% of CLAIM cards showed this misconception, build a brief 5-minute re-teach into the opening of Lesson 4 before introducing diffraction. (3) Were learners able to draw the normal correctly as perpendicular to the surface (not always vertical)? This is the most common diagrammatic error in wave reflection — if prevalent, prepare a short demonstration using a mirror and torch at the start of Lesson 4. (4) Did the DQB NEW QUESTION cards surface the word 'bend' or 'around' in relation to sound — indicating readiness for diffraction? Identify 2–3 learners whose cards already anticipate diffraction and plan to cold-call them first in Lesson 4's Predict phase to model high-quality scientific thinking for peers. (5) How equitably were roles distributed in the echo experiment — did female learners take on the active roles of Clapper and Timer, or default to Recorder? Adjust group role assignments in Lesson 4 to ensure equitable participation.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners stood at measured distances (20–40 m) from the school perimeter wall, clapped rhythmically, and timed echoes to calculate the speed of sound. Groups pooled their results on a class data board and compared calculated values (expected range: 300–380 m/s) to the accepted value of ~340 m/s.
What did I learn?	Sound reflects off hard surfaces; the angle of incidence equals the angle of reflection (law of reflection). Echoes are reflected sound waves. Reflection can explain why the fisherman hears some thunder from the direction of the cliff — but it cannot explain sound reaching the completely shadowed cove behind the headland.
How does this explain the phenomenon?	The fisherman's experience of hearing thunder from the direction of the rocky cliff can be partly explained by the reflection of sound waves off the hard rock surface, redirecting some of the sound energy toward him. However, reflection requires a line-of-sight path to the reflecting surface and cannot explain why sound reaches the protected cove with no reflective surface facing the storm — that part of the puzzle remains unsolved and points toward a different wave behaviour (diffraction) to be explored in Lesson 4.

LESSON 4 (80 minutes (2 × 40-minute lessons)): Refraction of Waves — Why Does the Water Bend at the Shore and the Radio Fade in the Rift Valley?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To enable learners to explain the refraction of waves using the change in wave speed at a boundary, apply the wave equation to calculate new wavelength when speed changes, and connect the phenomenon of refraction to real-life Kenyan contexts including coastal-swell bending near Mombasa, temperature-inversion effects on sound over Lake Victoria, and FM-radio fading in the Rift Valley.
Knowledge	Learners will know that: (i) refraction is the change in direction of a wave when it crosses a boundary between two media in which it travels at different speeds; (ii) frequency remains constant during refraction while wavelength changes; (iii) a wave bends towards the normal when it slows down and away from the normal when it speeds up; (iv) the wave equation $v = f\lambda$ can be used to calculate the new wavelength after a speed change at a boundary; (v) temperature gradients in air cause sound to refract, explaining why sound travels farther at night over Lake Victoria.
Skills	Learners will be able to: (i) draw and label a refraction diagram showing the incident ray, refracted ray, normal, and boundary; (ii) apply the wave equation $v = f\lambda$ to calculate wavelength when wave speed changes at a boundary while frequency remains constant; (iii) interpret video evidence of water-wave refraction in a ripple tank and connect it to coastal swell footage from Mombasa; (iv) construct a written explanation linking temperature inversion to downward refraction of sound over the lake; (v) update the Driving Question Board with a new, evidence-backed claim about wave bending.
Attitudes	Learners will appreciate that: (i) wave refraction is not an abstract concept but a phenomenon that shapes daily Kenyan experiences — from the fisherman's ability to hear a storm, to the matatu passenger's fading radio signal in the Rift Valley; (ii) scientific understanding of wave behaviour can inform practical decisions that protect communities and improve technology; (iii) careful observation of local phenomena (sound over a lake at night, coastal swells) is a valid starting point for scientific inquiry.
Key Inquiry Question	How are waves applicable in our day-to-day life? / How do waves affect our environment?
Purpose in Storyline	Lessons 1–3 established what waves are, introduced the wave equation, and explored reflection. The anchoring phenomenon showed that water waves change direction as they cross into the shallow bay near Kisumu, and that sound bends around the rocky headland. Learners could not yet explain the direction change. Lesson 4 now provides the mechanism — refraction — giving learners the causal story for the shallow-water bending they observed in the phenomenon, and extending it to explain the Lamu fishermen's night-time sound and the Rift Valley radio fade. This evidence directly advances the Driving Question Board entry on 'why water waves change direction at the shallow edge of the bay.'
Safety Notes	No hazardous materials are used in this lesson. If a jiko (charcoal stove) demonstration is used for the warm-air prediction, ensure it is placed on a stable surface away from learners, the room is ventilated, and only the teacher handles the stove. Learners must not lean over the jiko. If the lesson is conducted offline on ARES tablets, ensure devices are charged and screen brightness is managed to avoid eye strain during the video observation phase.

B. LESSON OVERVIEW

Lesson 4 places refraction at the centre of the Waves storyline. Learners begin by predicting what happens to the speed and direction of sound as it moves from cool air into the warm air rising above a jiko heating a pot of ugali — a tangible Kenyan entry point into the concept of wave speed changing with medium. They then observe two video clips embedded in the ARES platform: a ripple-tank slow-motion clip showing water waves bending as they cross from deep to shallow water, and a real-world

clip of coastal swell waves visibly refracting as they approach the Mombasa shoreline. Using these clips, learners construct a labelled refraction diagram and establish the rule: slower medium → wave bends towards the normal. In the Explain phase, learners apply this rule to two Kenyan sound contexts — the Lamu fishermen's experience of sound travelling farther at night due to temperature inversion, and the matatu FM-radio fading behind Rift Valley ridges. A structured numerical problem requires learners to apply $v = f\lambda$ to calculate the new wavelength when wave speed drops from 340 m/s to 320 m/s at an air-temperature boundary, with frequency held constant at 680 Hz. The lesson closes with a Driving Question Board update in which learners revise their prior DQB entry on shallow-water bending, and a cumulative model revision in which the refraction arrow is added to each learner's wave-behaviour diagram.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Phase 1 — PREDICT (15 minutes) VIDEO: Video: Bending Light Source: PhET Interactive Simulations (English) Search ARES for similar videos HTML: Refraction (Flexbook) Source: CK-12 Search ARES for similar readings	Learners are shown a photograph (displayed on the ARES platform) of a jiko with a sufuria of ugali cooking on it, with visible heat shimmer rising above it. They are asked two written predict questions in their exercise books: (1) 'Sound travels at about 340 m/s in cool air. Do you think it travels faster or slower in the warm air above the jiko — and why?' (2) 'If a sound wave is travelling through cool air and crosses into a layer of warm air at an angle, predict whether it will bend towards the warm-air side or away from it — draw a sketch.' Learners work individually for 4 minutes, then share with a shoulder partner for 2 minutes. The teacher takes three cold-call responses per question, recording key prediction words on the board under the headings FASTER / SLOWER and BENDS TOWARD / BENDS AWAY. The teacher does NOT confirm or correct — predictions stay visible on the board for the entire lesson so learners can return to them.	Display the jiko photograph and read the two predict questions aloud. Say: 'You have 4 minutes to write your own prediction — no right or wrong answers yet. Think about what you already know from Lessons 1 to 3 about wave speed and medium.' START the timer visibly. After 4 minutes say: 'Share your sketch with your shoulder partner. You have 2 minutes.' After partner share, cold-call 3 learners for Question 1 using the class register (rows 1, 5, 9). Ask each: 'What is your prediction and what evidence from our earlier lessons made you think that?' Allow 10 seconds WAIT TIME before prompting. Cold-call 3 different learners for Question 2. Record key words on the board verbatim — do not evaluate. Say: 'These predictions will be our test — by the end of today you will know who was right and why.' Leave the board visible. Transition: 'Now let us gather some evidence.'	Predict-before-observe (eliciting prior conceptions): By committing predictions to paper and sharing them publicly before any instruction, learners activate prior knowledge about wave speed and medium (built in Lessons 1–3). The visible prediction record on the board creates cognitive dissonance when observations contradict predictions, which research shows deepens encoding of the correct concept. Connecting the prediction to the everyday jiko makes the abstract concept of medium-dependent wave speed personally relevant from the first minute.	Teacher circulates during individual writing and reads over shoulders. Specific indicator: Can the learner articulate a reason for their speed prediction that references the medium (e.g., 'warmer air has less dense molecules so sound moves faster')? Can the learner sketch even a rough diagram showing a wave crossing a boundary? Learners who leave the diagram blank are noted for targeted support in the Observe phase. No marks are given; teacher uses a mental checklist of 5–6 learners to monitor during the lesson.
Phase 2 — OBSERVE (20 minutes) VIDEO: Video: Bending	Learners open the ARES media player and watch Clip A (ripple-tank, 2 min 30 s): slow-motion water waves in a ripple tank crossing from a deep region into a	Before Clip A: 'As you watch, do not write — just look. Ask yourself: what changes when the wave crosses from the deep side to the shallow side?' Play Clip A once.	Two-clip evidence sequence (concrete-to-real-world transfer): Clip A provides controlled, clear visual evidence in a simplified system (ripple tank) where	Observation table: Teacher checks that column (iv) reads 'towards the normal' for slowing waves. Specific error to watch for: learners who write 'away from the normal' (a

Light Source: PhET Interactive Simulations (English) Search ARES for similar videos HTML: Refraction (Flexbook) Source: CK-12 Search ARES for similar readings	shallow region marked by a submerged glass plate. The clip shows the wavefronts clearly compressing and bending as they cross the boundary. Learners watch once without writing. They then watch a second time and complete an ARES structured observation table with four columns: (i) What I see happening to the wave SPEED in the shallow region, (ii) What I see happening to the WAVELENGTH (gap between wavefronts), (iii) What I see happening to the DIRECTION of the wave, (iv) Does the wave bend TOWARDS or AWAY from the normal? Learners then watch Clip B (Mombasa coastal swell, 1 min 45 s): aerial footage of ocean swell waves visibly bending as they approach the shallow reef shelf near Mombasa, spreading around a headland. After Clip B, learners answer: 'How is what you see at the Mombasa coast the same as what you saw in the ripple tank?' Learners then, individually, draw a full refraction diagram in their books: a horizontal boundary line, a normal (dashed perpendicular), an incident ray above the boundary with wavefronts, and a refracted ray below the boundary with compressed wavefronts, labelling all five features.	'Now watch again and fill in your observation table.' Play Clip A a second time. Give 3 minutes for table completion. Cold-call 2 learners to read their column (iii) and (iv) entries. Allow 10 seconds WAIT TIME. Say: 'Interesting — hold that thought.' Before Clip B: 'This is the actual coast of Mombasa. You are going to see the same physics happening at a much larger scale. Watch for the moment the waves hit the shallow reef shelf.' Play Clip B once. Ask: 'Turn to your partner — 30 seconds — tell them one thing that is the same between the ripple tank and Mombasa.' Cold-call 2 pairs. After responses, say: 'Now draw the refraction diagram. Use the example structure on the ARES screen as your guide, but draw it yourself in your exercise book.' Display the diagram scaffold (boundary, normal, incident ray label prompts) on ARES. Circulate and give targeted verbal feedback: 'Is your normal perpendicular to the boundary? Which side of the normal is your refracted ray on?' Do NOT draw the diagram for learners. After 5 minutes, display a completed teacher model and ask learners to self-check their own diagrams.	variables are visible. Clip B immediately demands transfer to a real Kenyan coastal context, preventing the concept from remaining abstract. The structured observation table is a disciplined noticing tool — it forces learners to separate three distinct observable variables (speed, wavelength, direction) that they might otherwise conflate. The diagram construction task converts passive viewing into active representational practice, a core NGSS Science and Engineering Practice (developing and using models).	common misconception — confusing refraction with the reflection rule). Refraction diagram: All five labels must be present (incident ray, refracted ray, normal, boundary, wavefronts). The refracted ray must be on the correct side of the normal. Teacher collects 6 books at random at end of phase for rapid scan — returns before Explain phase begins.
Phase 3 — EXPLAIN (25 minutes) VIDEO: Video: Bending Light Source: PhET Interactive Simulations (English) Search ARES	The teacher introduces the formal vocabulary: refraction, angle of incidence, angle of refraction, normal, boundary. Learners add these to their diagrams with arrows. The class then works through two Kenyan context discussions in groups of four, followed by a numerical problem. CONTEXT 1 — Lamu fishermen:	Introduce vocabulary by pointing to the learners' own diagrams: 'You already drew these — now we are giving the parts their scientific names.' Do NOT re-draw — point to learner diagrams projected on screen. For Context 1: 'You have 5 minutes in your group of four. One person draws, one person labels, one person writes the explanation,	Structured context-transfer discussion (applying concepts to Kenyan phenomena): By requiring learners to construct their own diagram for the Lamu temperature-inversion scenario, they must actively apply the refraction rule (slow → bend toward normal; fast → bend away from normal) rather than passively receive it. The Rift	Numerical problem: (a) $\lambda = 340/680 = 0.50$ m. (b) $\lambda = 360/680 \approx 0.529$ m. (c) The wave bends away from the normal because it speeds up. Teacher checks for three specific errors: (1) learner changes frequency instead of wavelength; (2) learner uses incorrect formula rearrangement; (3) learner states 'bends toward

for similar videos  HTML: Refraction (Flexbook) Source: CK-12  Search ARES for similar readings	Learners are given a short ARES reading card (150 words): 'At night, the air near the surface of the sea is cooler than the air higher up — this is called a temperature inversion. Sound travels faster in warm air. At night, sound waves travelling upward hit the warmer upper layer and refract back downward toward the water surface. The sound bounces between the sea surface and the warm upper air layer, travelling much farther than it would during the day. This is why Lamu fishermen have long known that voices and drumbeats carry much farther across the water at night.' Groups discuss: 'Draw a diagram showing how this temperature inversion refracts the sound downward. Label the cool air, the warm air layer, the sound wave path, and explain why the wave bends downward.' CONTEXT 2 — Rift Valley matatu radio: 'A matatu travelling from Nairobi to Nakuru loses its FM radio signal as it passes behind a ridge near Mai Mahiu. The FM waves (which travel in straight lines in uniform air) cannot bend around the solid ridge the way sound bends around a temperature layer — they need a different mechanism. But the signal also fades gradually as the matatu descends into the valley — the atmosphere refracts the signal slightly. Why does the signal fade completely behind the ridge, and only partly refract in the valley?' Groups discuss for 4 minutes, then report out. NUMERICAL PROBLEM: Displayed on ARES. 'A sound wave travels through cool air at 340 m/s with a frequency of 680 Hz. It crosses into a warmer air layer where its speed increases	one person will present. Switch roles from last lesson.' After 5 minutes, cold-call one group to present their diagram. Ask the presenting group: 'Why does the wave bend downward and not upward?' Allow 10 seconds WAIT TIME. If no response after 10 seconds, prompt: 'What does the wave do when it enters a region where it travels faster — does it bend toward or away from the normal?' For Context 2: After 4-minute group discussion, cold-call a different group. Ask: 'What is the difference between how sound refracts around the temperature layer and why the FM signal cannot do the same thing behind a solid ridge?' Allow 15 seconds WAIT TIME — this is a higher-order question. Accept partial answers and build: 'You are right that the ridge is solid — we will explore diffraction in Lesson 5 to finish this explanation. Add it to your DQB question list.' For the numerical problem: After individual work, ask: 'Who got the same answer for both (a) and (b)? Hands up.' If many hands, cold-call one to explain. If few, work through step (a) together on the board using the wave equation $v = f\lambda$, asking learners to call out each substitution: 'What is v? What is f? So $\lambda = ?$' Accept shouted answers for substitution steps only — require written working for the final answer. Emphasise: 'Frequency does not change at the boundary — only speed and wavelength change. Write that rule in a box in your notes.'	Valley matatu context deliberately creates productive confusion — the FM signal fade requires diffraction, not refraction — which seeds the DQB question for Lesson 5 and demonstrates that one phenomenon can require multiple wave behaviours to fully explain. The numerical problem is an NGSS Mathematical and Computational Thinking practice: learners apply a quantitative relationship ($v = f\lambda$) to a boundary condition, reinforcing that frequency is the conserved quantity while speed and wavelength co-vary.	the normal' when speed increases. Teacher records names of learners making error (1) — this is the most persistent misconception in refraction and must be addressed before Lesson 5. Context diagrams are checked for correct direction of bending (downward toward water surface for temperature inversion). Exit indicator: every learner must be able to state the rule 'when a wave slows down it bends toward the normal' in their own words.
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	to 360 m/s. The frequency does not change. (a) Calculate the wavelength in the cool air. (b) Calculate the wavelength in the warm air. (c) Does the wave bend towards or away from the normal as it enters the warm air? Explain.' Learners work individually for 6 minutes, then compare with a partner.			
Phase 4 — DRIVING QUESTION BOARD (DQB) UPDATE (10 minutes)  VIDEO: Video: Bending Light Source: PhET Interactive Simulations (English)  Search ARES for similar videos  HTML: Refraction (Flexbook) Source: CK-12  Search ARES for similar readings	Learners return to the class Driving Question Board (a shared physical or digital space maintained across all 8 lessons). They locate the card/sticky note placed in Lesson 1 or 2 that asked: 'Why do the water waves in the bay change direction as they reach the shallow water near the shore?' Learners now write a new CLAIM card in a different colour: 'The water waves refract (change direction) because they slow down as the water gets shallower. When the wave slows down, it bends toward the normal at the boundary between deep and shallow water. This is called refraction.' Learners also add a new QUESTION card for anything Lesson 4 could not yet explain: 'Why does the FM signal disappear completely behind the Rift Valley ridge — refraction alone does not explain this. What else can waves do?' Each learner updates their personal DQB tracker (a running table in their exercise book with columns: Lesson New evidence What it explains about the phenomenon New questions it raises).	Direct learners to the DQB: 'Find the question card about the shallow-water bending. We have been carrying that question since Lesson 1. Today we can finally answer it.' Give 3 minutes for learners to write their claim card. Cold-call one learner to read their claim aloud. Ask the class: 'Does this claim match the evidence we gathered today? Thumbs up if yes, thumbs sideways if you are not sure.' Address any thumbs-sideways by asking that learner: 'What part of the evidence are you not sure about?' Allow 10 seconds WAIT TIME. Then direct learners to their new question card about the Rift Valley ridge: 'We know refraction cannot explain why the signal disappears completely behind a solid ridge. Write that as our new question. We will answer it in Lesson 5.' Update the DQB together. Say: 'Notice that every lesson we answer one question and create a new one — this is exactly how science works.'	Claim-Evidence-Reasoning (CER) on the DQB: The DQB is not a static display — it is a cumulative argument board. By requiring learners to physically replace an open question with a reasoned claim, they experience the progression of scientific understanding. The colour-coded cards (question = one colour, claim = another) make the growing explanatory power of the unit visually trackable. The deliberate planting of a new, unresolved question (ridge diffraction) maintains inquiry momentum and prevents premature closure — a key feature of NGSS storyline design.	Learner DQB tracker tables are checked for logical coherence: Does the 'What it explains' column correctly reference refraction and the speed change? Does the 'New questions' column show genuine extension of thinking (not just a restatement of old questions)? Teacher reads 5 tracker tables during the final phase. Specific indicator: The learner can distinguish what refraction explains (direction change at boundary due to speed change) from what it cannot explain (disappearance of signal behind a solid obstacle), demonstrating boundary awareness of the concept.
Phase 5 — MODEL BUILDING (10 minutes)  VIDEO:	Learners retrieve their cumulative wave-behaviour model — a diagram they have been building since Lesson 1 showing the Lake Victoria fisherman phenomenon	Display the original phenomenon image (Lake Victoria fisherman, storm, headland, shallow bay). 'Open your cumulative model. You added reflection in Lesson 3.	Cumulative model revision (developing and using models — NGSS SEP 2): The iterative, lesson-by-lesson model build is the spine of the storyline. Each	Model check: The refraction addition must show (i) a boundary between deep and shallow water (or between cool and warm air), (ii) wavefronts that compress on the

Video: Bending Light Source: PhET Interactive Simulations (English) Search ARES for similar videos HTML: Refraction (Flexbook) Source: CK-12 Search ARES for similar readings	with all wave behaviours annotated. In Lessons 2–3 they added reflection arrows (sound bouncing off the cliff). Today they add a refraction panel: a new section of the diagram showing the water wave crossing from deep lake water into the shallow bay, with a clear boundary line, normal, incident wavefront, and refracted wavefront bending toward the shore. They also add a second refraction annotation to the sound path: showing how sound from the distant storm refracts downward at night due to the temperature inversion, following a curved path that reaches the fisherman around the headland. Learners write one sentence beneath the refraction addition: 'This explains why the water waves change direction at the shallow edge — they slow down and bend toward the normal, just like in the ripple tank.' They share their updated model with their shoulder partner and identify one similarity and one difference between their two models.	Today add refraction in two places: the water wave at the shallow boundary, and the sound wave bending in the temperature inversion at night.' Give 6 minutes for drawing. Circulate and give specific feedback: 'Is your refracted wavefront closer together than your incident wavefront? It should be — wavelength decreases when the wave slows down.' After 6 minutes: 'Share with your shoulder partner. Tell them: one thing your models have in common, and one thing that is different.' Allow 2 minutes for partner share. Cold-call one pair: 'What was the most important difference between your models?' Allow 10 seconds WAIT TIME. Close by saying: 'Your model is now explaining two wave behaviours — reflection and refraction. Next lesson we add a third — diffraction — which will finally explain how sound gets completely around the solid headland and into the cove.'	addition makes the model more explanatorily powerful while remaining anchored to the same phenomenon. By requiring learners to add refraction to both the water wave (visible, spatial) and the sound wave (invisible, temporal — night vs. day), they practise applying one mechanism across two different wave types, deepening generalisation. The partner comparison step surfaces representational differences that reveal conceptual differences — a diagnostic opportunity for the teacher without requiring formal assessment.	slow side, (iii) the wave bending toward the normal on the slow side. Teacher checks that the explanatory sentence references 'slowing down' as the cause of bending — not simply 'hitting the boundary.' Any learner whose sentence reads 'the wave bends because it hits the shallow water' (without referencing speed change) is flagged for a brief targeted question at the start of Lesson 5: 'What specifically causes the bending — is it the boundary itself or what happens to the wave's speed at the boundary?'
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record brief notes in your professional journal:

1. **PREDICTION QUALITY:** Did learners' initial predictions about warm-air speed reveal useful prior knowledge, or did they reveal misconceptions about sound and temperature that need revisiting? Were there learners who predicted 'slower in warm air' — and if so, how did you address this without shaming them during the lesson?
2. **OBSERVATION DEPTH:** When learners watched Clip A (ripple tank), did they notice all three changes (speed, wavelength, direction), or did they focus only on direction? If wavelength changes were missed, consider adding a freeze-frame pause in a future lesson delivery — pause the clip at the moment the wave crosses the boundary and ask: 'Count the gaps between the wavefronts on each side. What do you notice?'
3. **NUMERICAL PROBLEM:** How many learners made Error (1) — changing frequency instead of wavelength? If more than 30% of the class made this error, open Lesson 5 with a 5-minute re-teach of the wave equation, explicitly contrasting frequency (property of the source, does not change at boundary) with wavelength (property of the medium, changes with speed).
4. **DQB QUALITY:** Were learners' claim cards sufficiently causal (referencing speed change as the mechanism) or were they descriptive only ('the wave bends')? Strong claims include a because statement. In future, model a complete CER claim aloud before asking learners to write their own.
5. **KENYAN CONTEXT RESONANCE:** Did learners engage more with the Lamu fishermen context or the Rift Valley matatu radio context? Note which context generated richer discussion — use that as the opening hook for Lesson 5's diffraction phase. If learners from the Rift Valley region are in your class, invite them to share

personal experience of radio signal loss — lived experience is powerful evidence.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners watched slow-motion ripple-tank footage (Clip A) showing water wavefronts compressing and bending as they crossed from a deep region into a shallow region, and then Mombasa coastal-swell footage (Clip B) showing the same behaviour at the scale of the Indian Ocean. They completed a structured observation table recording changes in wave speed, wavelength, and direction at the boundary, and drew a labelled refraction diagram (incident ray, refracted ray, normal, boundary, wavefronts).
What did I learn?	Learners learned that refraction is the change in direction of a wave caused by a change in its speed when it crosses a boundary between two media. The wave bends toward the normal when it slows down and away from the normal when it speeds up. Frequency is conserved at the boundary; wavelength changes in proportion to speed ($v = f\lambda$). A temperature inversion at night causes sound to travel faster in the upper warm layer and refract back downward, explaining why Lamu fishermen hear voices farther across the water at night. Applying $v = f\lambda$: at 680 Hz, a wave travelling at 340 m/s has wavelength 0.50 m; at 360 m/s it has wavelength ≈ 0.529 m.
How does this explain the phenomenon?	Refraction explains why the water waves in the Lake Victoria phenomenon change direction as they cross from the deep open lake into the shallow bay near the Kisumu shore — the wave slows down in the shallower water and bends toward the normal at the deep-to-shallow boundary. This is the same mechanism seen in the Mombasa coastal swell. Refraction also partially explains how sound from the distant storm reaches the fisherman at night: a temperature inversion refracts the sound downward, allowing it to travel farther along the water surface. However, refraction alone cannot explain why sound reaches the protected cove behind the solid headland, or why the FM radio signal disappears completely behind a Rift Valley ridge — those require a third wave behaviour (diffraction), which will be the focus of Lesson 5.

LESSON 5 (80 minutes (2 × 40-minute lessons)): Diffraction of Waves — Sound Around the Corner and Signals Over the Hill

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To demonstrate, explain, and apply the diffraction of waves — including sound waves bending around obstacles and gaps — connecting this phenomenon to the anchoring story of the Kisumu fisherman and to real Kenyan technology contexts (mobile signals in the Rift Valley, radio broadcasts between Nairobi and Kisumu).
Knowledge	Learners explain that diffraction is the spreading of waves around obstacles or through gaps; that the degree of diffraction depends on the relationship between wavelength and gap/obstacle size; that long wavelengths (e.g., sound, FM radio) diffract more readily than short wavelengths (e.g., light); and that diffraction explains why sound is audible around corners and why mobile/radio signals can reach behind hills.
Skills	Learners carry out an experiment to establish that sound waves can be heard around a corner (KICD: 'carry out experiments to establish that sound waves can be heard round a corner'); construct ray/wavefront diagrams showing diffraction through narrow and wide gaps; interpret simulation evidence of water-wave diffraction; and calculate wave characteristics using the wave equation.
Attitudes	Learners appreciate the applications of waves in real life (KICD); demonstrate unity by taking assigned roles during group experiments; and show responsibility by handling apparatus carefully and recording results honestly.
Key Inquiry Question	How are waves applicable in our day-to-day life? How do waves affect our environment?
Purpose in Storyline	In Lessons 1–2 learners established wave properties and the wave equation. Lesson 3 explored reflection (echo = the fisherman hearing thunder bounce off the cliff). Lesson 4 explored refraction (water waves bending as they cross from deep lake into the shallow bay). Lesson 5 now completes the anchoring phenomenon: diffraction explains how sound reaches the fisherman even when the storm is hidden behind the rocky headland, and why mobile-phone signals still reach a caller behind a hill in the Rift Valley. By the end of this lesson learners can explain all three behaviours (reflection, refraction, diffraction) in learner-generated language on the Driving Question Board, advancing the unit's central question.
Safety Notes	Learners must not shout or clap loudly near others' ears at close range — maintain a minimum 1-metre separation between the sound-maker and the nearest listener during the doorway experiment. No running in corridors. All learners who are not actively making sound should remain silent so results are valid. If a signal bell or clapper is used, inspect it for sharp edges before use.

B. LESSON OVERVIEW

In this 80-minute double lesson, learners investigate diffraction — the spreading of waves around obstacles and through gaps. The lesson opens with a predict task about whether a speaker's voice is audible around a doorway corner without a line of sight. Learners then conduct a classroom doorway experiment (sound-around-a-corner) and analyse a water-wave diffraction simulation, generating qualitative evidence. In the explain phase they construct wavefront diagrams for narrow and wide gaps and connect diffraction to the anchoring phenomenon (sound reaching the fisherman around the rocky headland) and to Kenyan technology (Rift Valley mobile signals; Nairobi–Kisumu radio coverage). The Driving Question Board is updated to show that all three wave behaviours — reflection, refraction, and diffraction — together explain the full fisherman phenomenon. The lesson closes with a cumulative model revision in which learners add diffraction arrows to their wavefront diagram of the Kisumu bay scene.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Phase 1 — Predict  VIDEO: Video: Bending Light Source: PhET Interactive Simulations (English)  Search ARES for similar videos  HTML: Diffraction (Flexbook) Source: CK-12  Search ARES for similar readings	Each learner receives a half-page 'Predict Card' showing a top-down diagram of a classroom corridor with an open doorway. A speaker (labelled 'Akinyi') stands just outside the doorway; a listener (labelled 'Kamau') stands around the corner in the corridor, completely out of the line of sight. The card asks three questions: (1) Will Kamau be able to hear Akinyi's voice clearly, faintly, or not at all? Draw your prediction on the diagram using arrows or wavefronts. (2) Will Kamau be able to see Akinyi? Explain why or why not. (3) If Akinyi moves further from the doorway so that the gap between her and the wall is larger, will the sound reach Kamau more easily or less easily? Write one sentence of reasoning. Learners work individually for 4 minutes in silence, then share predictions with a shoulder partner for 2 minutes. The teacher records a tally of class predictions on the board (columns: 'Will hear clearly / Will hear faintly / Will not hear') using a show-of-hands cold call across the whole class.	Distribute Predict Cards face-down; say: 'Do not turn over until I say go — this is your thinking, not your neighbour's.' Start a 4-minute silent timer visible on the board. Circulate and note 3–4 students with interesting or divergent predictions (do not correct yet). After 4 minutes: 'Turn and share your prediction arrow diagram with your shoulder partner — you have 2 minutes.' After partner talk, conduct a whole-class cold-call tally: 'Hands up — who predicted Kamau will hear Akinyi clearly?' Count aloud and write the number. Repeat for 'faintly' and 'not at all'. Then cold-call 3 students — one from each group — asking: 'Josephine, tell me one reason for your prediction.' Allow 10–12 seconds WAIT TIME after each question before accepting the answer. Write key prediction reasons on the board verbatim (e.g., 'sound needs a straight path', 'sound bounces off walls', 'sound bends'). Do NOT affirm or correct yet. Say: 'Keep your Predict Card — we will revisit these in 20 minutes.'	Elicit–Confront–Resolve (Predict phase only: Elicit). By asking learners to draw arrows/wavefronts before any instruction, the teacher surfaces prior conceptions — most learners will predict 'no sound' because they equate wave travel with ray optics (straight-line thinking). This productive misconception becomes the cognitive hook that the Observe phase will confront, making the evidence surprising and therefore memorable.	Observable indicator: Learner's Predict Card shows a drawn prediction (arrows or wavefronts) and at least one written sentence of reasoning — even if incorrect. Teacher scans cards during circulation and identifies learners who draw straight-line 'rays' stopping at the wall (most common) vs. those who draw curved spreading lines (rare, indicating prior knowledge). These two groups are mentally noted for targeted questioning in the Explain phase. Tally counts on the board give the teacher a class-level snapshot of the dominant misconception.
Phase 2 — Observe  VIDEO: Video: Bending Light Source: PhET Interactive Simulations (English)  Search ARES for similar videos  HTML:	PART A — Doorway Sound Experiment (25 minutes). The class divides into groups of 5. Each group takes a turn at the open classroom doorway; the rest observe and record. Within each group: one learner (the 'speaker') stands 0.5 m outside the doorway and claps twice or says a short phrase ('Habari za asubuhi!') at normal conversational volume; two learners (listeners L1 and L2)	Before Part A: Demonstrate the setup at the doorway with one group while all others watch. Say clearly: 'The speaker stands here — outside. Listeners L1 and L2 stand here and here — around the corner, no line of sight to the speaker. L3 stands here — direct line of sight, same distance. This is our control.' Ask the class: 'Why do we need L3?' — WAIT 10 seconds — cold-call 2 students. Establish	Evidence-to-Claim mapping. Learners generate two parallel data sets — audibility ratings from the real doorway experiment and sketch diagrams from the simulation — and are then asked to identify a common pattern across both. This cross-referencing strategy builds the understanding that diffraction is a property of all waves (sound and water), not a special case,	Observable indicators: (1) Class results table on the board is complete — all groups have entered audibility ratings for all three gap conditions, showing a trend of increased spreading at smaller gaps. Teacher checks for outliers and discusses them. (2) Each learner's exercise book contains three annotated wavefront sketches from the simulation — teacher does a quick

Diffraction (Flexbook) Source: CK-12 Search ARES for similar readings	stand around the corner in the corridor at 1 m and 2 m from the doorway edge respectively, both out of the speaker's line of sight; one learner acts as the 'control listener' (L3) who stands in the direct line of sight of the speaker at 1 m, confirming sound is audible in the direct path; one learner records results in a class results table on the board. Each group tests three conditions: (i) doorway fully open (gap $\approx$ 1 m wide); (ii) doorway half-blocked with a book or bag (gap $\approx$ 0.5 m); (iii) doorway almost fully blocked (gap $\approx$ 0.1 m, leaving a narrow slit). Listeners rate audibility as 3 (clear), 2 (faint), or 1 (cannot hear). Results are compiled in a shared class table. After all groups rotate, learners write two observations in their exercise books: 'What happened to the sound around the corner?' and 'What happened as the gap got narrower?' PART B — Water-Wave Diffraction Simulation (10 minutes). Using the ARES offline simulation player, learners watch a 3-minute animation of water waves passing through gaps of decreasing width (wide gap $\rightarrow$ medium gap $\rightarrow$ narrow gap $\approx$ wavelength). Learners sketch three diagrams in their books: wavefronts before the gap (parallel lines) and wavefronts after the gap for each of the three conditions. They annotate each sketch with one observation sentence.	that L3 confirms sound is being produced, not that the room is just echoing. Manage the rotation so each group takes no more than 4 minutes at the doorway. While groups rotate, circulate to the results table and prompt: 'Wanjiku, what pattern do you notice across the three gap sizes?' During Part B (simulation): Project the simulation on the screen or ensure each pair has access on a tablet. Pause the simulation after the wide-gap frame and ask: 'Look at the wavefronts after the gap — what shape are they?' WAIT 12 seconds. Cold-call 3 students. Pause again after the narrow-gap frame: 'What changed? Use the word spread in your answer.' WAIT 10 seconds. Cold-call 2 students. After Part B, instruct: 'In your exercise book, write two sentences starting with: I observed... and I wonder...'	because both data sets show the same trend: spreading increases when gap width approaches wavelength. This directly confronts the Predict-phase misconception that sound travels only in straight lines.	'books open' scan during the transition to the Explain phase, looking specifically for curved wavefronts after the narrow gap. A learner who draws straight parallel lines after the narrow gap has not yet grasped diffraction and is flagged for targeted support.
Phase 3 — Explain  VIDEO: Video: Bending Light Source: PhET Interactive	Learners receive a structured diagram worksheet with two outline drawings: (A) a barrier with a wide gap (gap width $\gg$ wavelength) and (B) a barrier with a narrow gap (gap width $\approx$ wavelength). Using the simulation	Distribute the diagram worksheet. Say: 'Use your simulation sketches from Part B as your guide — do not start from memory.' Give 8 minutes for diagram drawing. Circulate and look for three things: (i) Are incoming wavefronts drawn	Diagram annotation with Phenomenon Connection. Drawing wavefront diagrams requires learners to externalise their internal model of wave spreading. The Phenomenon Connection boxes force them to translate the	Observable indicators: (1) Diagram B on the worksheet shows curved wavefronts spreading into the geometric shadow behind the barrier — this is the key indicator that the learner understands diffraction as spreading, not

Simulations (English) Search ARES for similar videos  HTML: Diffraction (Flexbook) Source: CK-12 Search ARES for similar readings	evidence and the doorway results, learners draw incoming parallel wavefronts and then draw the outgoing wavefronts after the gap, labelling: 'incident wavefronts', 'diffracted wavefronts', 'direction of energy spread', and 'shadow region'. They annotate diagram B with the word DIFFRACTION and write a definition in their own words below. Learners then complete a 'Phenomenon Connection' box on the same worksheet with three prompts: (1) 'The fisherman at Kisumu hears thunder around the rocky headland because...' (2) 'My mobile phone signal still works when I walk behind a hill near Nakuru/Kericho because...' (3) 'Radio broadcasts from Nairobi can be received in Kisumu even though the Earth curves because...'. Finally, learners solve two calculation problems using the wave equation ($v = f\lambda$): Problem 1 — An FM radio station in Nairobi broadcasts at 98.4 MHz. The speed of radio waves is 3×10^8 m/s. Calculate the wavelength and state whether this wave would diffract around a hill 200 m wide. Problem 2 — A sound wave has a frequency of 340 Hz and travels at 340 m/s. Calculate its wavelength and compare it to the doorway gap width used in the experiment.	as straight parallel lines? (ii) Are outgoing wavefronts after the narrow gap drawn as curves spreading into the shadow region? (iii) Is the shadow region labelled? After 8 minutes, cold-call one student to sketch their diagram B on the board. Ask the class: 'Thumbs up if yours looks similar, thumbs sideways if different.' Address any differences. Then direct attention to the Phenomenon Connection box. Say: 'Sentence 1 is about our fisherman in Kisumu. Think back to the very first lesson. The sound goes around the headland. Using the word diffraction, complete this sentence.' WAIT 12 seconds. Cold-call Omondi: 'Omondi, read your sentence.' Then ask: 'Who has a different ending — Fatuma?' For the calculation problems, say: 'You have already used $v = f\lambda$ in Lesson 2. Write the formula first, substitute, then solve — show all steps.' Give 6 minutes. Cold-call one student per problem to write working on the board. Ask the class: 'Does the wavelength of the FM radio wave (≈ 3 m) compare to a 200 m hill — would it diffract easily?' WAIT 10 seconds. Elicit the concept that significant diffraction requires gap/obstacle size $\approx$ wavelength, so the 3 m wave diffracts poorly around a 200 m hill but the diffraction of longer-wavelength (AM) radio waves is more significant.	abstract diagram into the specific Kisumu fishing-village context, the Rift Valley mobile signal, and the Nairobi–Kisumu radio link — moving from representation to application. The calculation problems anchor diffraction qualitatively to quantitative reasoning (wavelength vs. gap size), consolidating the wave equation from earlier lessons.	reflection or refraction. (2) Phenomenon Connection sentence 1 contains the word 'diffraction' and correctly identifies the headland gap as the diffracting structure. (3) Calculation working shows formula written first, correct substitution, and a correct wavelength value with units. Teacher collects worksheets at the end of the lesson for written feedback.
Phase 4 — Driving Question Board (DQB) Update  VIDEO: Video: Bending	Learners return to the class Driving Question Board — a large poster or whiteboard section that has been accumulating sticky notes since Lesson 1. The board has three columns labelled: 'What we observed', 'What we can now	Hand out sticky notes. Read the prompt aloud twice. Say: 'You have 3 minutes — one sticky note per person, one sentence. Start with Reflection because..., Refraction because..., Diffraction because...' Circulate and check	Cumulative Explanation Synthesis. The DQB sticky-note task requires learners to integrate three lessons of evidence (reflection, refraction, diffraction) into a single coherent narrative about the anchoring phenomenon. This is a	Observable indicator: The sticky note contains all three wave properties (reflection, refraction, diffraction) each linked to a specific part of the Kisumu fisherman phenomenon — not just to abstract definitions. A note that

Light Source: PhET Interactive Simulations (English) Search ARES for similar videos  HTML: Diffraction (Flexbook) Source: CK-12 Search ARES for similar readings	explain', and 'What we still wonder'. Each learner is given one sticky note. They must write one sentence that completes the following prompt (displayed on the board): 'The fisherman at Kisumu can now be explained by THREE wave behaviours: reflection because..., refraction because..., and diffraction because...' This is the first time learners are asked to integrate all three properties into a single explanation. After writing, volunteers place their sticky notes in the 'What we can now explain' column. The teacher then facilitates a 5-minute whole-class synthesis, reading aloud 3–4 sticky notes and identifying common language. The class agrees on a single 'consensus explanation' sentence that the teacher writes in bold on the DQB: 'When the storm struck over Lake Victoria near Kisumu, the thunder sound reflected off the cliff (echo), the water waves refracted as they entered the shallow bay (changing direction), and the sound diffracted around the rocky headland into the cove (spreading around the obstacle).'	that learners are writing all three properties (not just diffraction). After 3 minutes: 'Place your note in the Can now explain column.' Select 3 notes to read aloud — choose one strong, one partially correct, one with a common error. For the error note, ask the class: 'What needs to be adjusted here?' WAIT 10 seconds. Do NOT identify the author. Then write the consensus sentence on the DQB and say: 'This is our class explanation for the fisherman. Does anyone want to challenge any part of it?' WAIT 12 seconds. Accept challenges respectfully. Point to the 'Still wonder' column: 'What about mobile-phone signals and the internet — how do engineers use diffraction deliberately? That's a question for our next lesson on applications.'	sensemaking move because it prevents learners from treating each property as an isolated topic and instead builds a connected explanatory model — which is the goal of the entire evidence-gathering sequence.	says only 'diffraction is when waves spread' without connecting to the headland/cove does not yet meet the standard. Teacher uses the sticky notes as a quick formative check: if more than one-third of notes are missing one or more properties, the opening of the next lesson must revisit the integration before proceeding.
Phase 5 — Model Building  VIDEO: Video: Bending Light Source: PhET Interactive Simulations (English) Search ARES for similar videos  HTML: Diffraction (Flexbook)	Learners open their cumulative wavefront diagram of the Kisumu bay scene — started in Lesson 3 (reflection arrows) and updated in Lesson 4 (refraction bend at the shallow-water boundary). Today they add a third layer: diffraction around the rocky headland. Using a different colour (suggested: green for diffraction, to join blue for reflection and red for refraction used in previous lessons), learners draw: (1) The gap between the rocky headland and the shore as a narrow opening. (2) Incoming	Say: 'Open to your Kisumu bay diagram. You should see blue reflection arrows and red refraction curves from earlier lessons. Today we add green diffraction.' Hold up a teacher model diagram showing all three colours as a reference. Give 8 minutes for learners to add diffraction layer and inset diagram. Circulate and ask targeted questions: 'Chebet, which part of your diagram represents the headland gap — is it wide or narrow compared to the sound wavelength?' WAIT 10 seconds.	Progressive Model Revision. Each lesson in the sequence adds one explanatory layer to the same diagram, so the model grows in complexity alongside learner understanding. This strategy makes cumulative learning visible — learners can literally see three colours of explanation on a single page — and reinforces that science builds models iteratively from evidence rather than replacing one idea with another.	Observable indicator: The updated cumulative diagram contains all three colour-coded layers (reflection, refraction, diffraction) correctly placed in the Kisumu bay geography, with the diffraction layer showing curved/circular wavefronts spreading into the cove behind the headland. The inset Rift Valley diagram is present and labelled. A diagram that shows diffraction waves as straight lines, or that places diffraction at the wrong feature (e.g., at the shallow-water boundary instead of the

Source: CK-12 Search ARES for similar readings	wavefronts from the storm side. (3) Diffracted circular/curved wavefronts spreading into the protected cove. (4) A label: 'Diffraction — sound spreads around the headland into the cove'. They also add a small inset diagram in the corner of the page showing a cross-section of the Rift Valley hills with a mobile-phone tower on one side, a hill in the middle, and a phone user on the other side — with curved wavefronts bending around/over the hill and an annotation: 'FM/mobile signal diffracts around hill ($\lambda \approx 0.1\text{--}1\text{ m}$; hill gap $> \lambda$, so partial diffraction)'. The completed diagram now tells the full story of the anchoring phenomenon in a single visual model.	'Ochieng, your diffracted wavefronts are straight lines — what should they look like after passing through a narrow gap?' After 8 minutes, ask 2–3 learners to hold up their diagrams for a gallery-style peer review — the class responds with one 'glow' (something correct) and one 'grow' (something to improve) per diagram. Close the lesson by pointing to the full model: 'In Lessons 3, 4, and 5 we have built a complete scientific explanation for what the fisherman experienced. In Lesson 6 we will apply all three properties to real Kenyan problems — road safety mirrors, hospitals, and telecommunications towers.'	headland gap), indicates a residual misconception to address at the start of Lesson 6.
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D. TEACHER REFLECTION

After the lesson, consider the following questions: (1) During the Predict phase, what was the dominant misconception in this class — did most learners believe sound could not travel around the corner at all, or did they think it would bounce off the far wall? How did this affect the surprise value of the doorway experiment? (2) In the doorway experiment, did the audibility ratings across groups show a clear trend (louder/more audible at smaller gaps), or were results messy? If messy, what variables were uncontrolled (background noise, different vocal volumes, corridor acoustics) and how would you address them in a repeat? (3) Were learners able to connect the simulation wavefront diagrams to the real doorway result — i.e., did they see both as the same phenomenon? If not, what bridging question would help? (4) When completing the Phenomenon Connection box, how many learners spontaneously used the word 'wavelength' in their explanation of why mobile signals diffract around hills? This indicates whether the quantitative link (λ vs. gap size) from the calculation problems is being absorbed. (5) On the DQB sticky notes, how many learners successfully integrated all three wave properties (reflection, refraction, diffraction) into a single Kisumu fisherman explanation? If fewer than two-thirds achieved this, plan a 5-minute opening review at the start of Lesson 6 using a 'three-colour sorting' activity before moving to applications.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners carried out a classroom doorway experiment in which a speaker said 'Habari za asubuhi!' just outside an open doorway while listeners positioned around the corner (out of line of sight) rated audibility at three gap widths. They also watched and sketched a water-wave diffraction simulation showing wavefronts spreading through gaps of decreasing width.
What did I learn?	Learners constructed wavefront diagrams showing that waves spread (diffract) around obstacles and through gaps, and that diffraction is greatest when the gap width is close to the wavelength. They connected diffraction to why sound reaches the Kisumu fisherman around the rocky headland and why mobile/FM radio signals can be detected behind hills in the Rift Valley.
How does this explain the	Learners updated the class Driving Question Board with an integrated three-property explanation of the anchoring phenomenon

phenomenon?	(reflection = echo off cliff; refraction = water-wave bending at shallow bay; diffraction = sound spreading around headland into cove) and added a green-coded diffraction layer to their cumulative Kisumu bay wavefront diagram.
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LESSON 6 (80 minutes): Applications of Wave Properties in Real Life

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners move from understanding wave properties in isolation to seeing how reflection, refraction, and diffraction are deliberately harnessed in Kenyan technology and institutions — deepening the explanation of the anchoring phenomenon and equipping learners to answer the driving question with real-world evidence.
Knowledge	Learners will be able to: (a) identify and describe at least four real-life applications of wave properties (SONAR depth-sounding by KMFRI on Lake Victoria; ultrasound scanning at Kenyatta National Hospital; cats-eye road reflectors and speed cameras on the Nairobi–Mombasa highway; seismic wave monitoring by the Kenya Meteorological Department); (b) link each application explicitly to one or more of the wave properties — reflection, refraction, or diffraction — studied in Lessons 3–5; (c) explain how the behaviour of waves in the anchoring phenomenon (sound bending around the rocky headland at Lake Victoria; water waves refracting into the shallow bay) is the same physics that underpins each technological application.
Skills	Learners will: (a) research an assigned application using print or digital media and synthesise findings into a poster or digital slide; (b) present findings coherently to peers in a structured 3-minute group presentation; (c) formulate and respond to peer questions using scientific language; (d) apply the wave equation and wave-property vocabulary to explain each application; (e) update the Driving Question Board by placing application cards beside corresponding wave-property headers.
Attitudes	Learners will: (a) appreciate how scientific knowledge of waves directly improves safety, health, and economic outcomes for Kenyan communities; (b) value collaboration and equitable contribution within research groups; (c) demonstrate respect for peers by listening actively and asking constructive questions during presentations; (d) show curiosity about further technological applications of waves in Kenya and globally.
Key Inquiry Question	How are waves applicable in our day-to-day life? How do waves affect our environment?
Purpose in Storyline	Lessons 1–5 built the evidence base: learners defined wave terms, used the wave equation, and investigated reflection, refraction, and diffraction through experiments and simulations. Lesson 6 is the Explain/Apply pivot — learners now deploy that evidence to show that the puzzling observations of the fisherman at Lake Victoria (sound bending around the headland, water waves refracting into the bay, mobile signals fading behind Rift Valley hills) are not curiosities but are the same physics engineered into sonar vessels on Lake Victoria, ultrasound machines at KNH, cats-eye reflectors on the Nairobi–Mombasa highway, and seismographs at KMD. Lesson 7 will use these applications as a platform for evaluating wave effects on the environment; Lesson 8 will produce the final cumulative model.
Safety Notes	No hazardous materials are used. If learners use personal digital devices for research, remind them to apply cyber-safety practices (access only approved educational sources; do not share personal information online). Ensure poster-making materials (scissors, rulers, markers) are handled responsibly. If the class moves to a computer lab, follow the school's ICT acceptable-use policy.

B. LESSON OVERVIEW

Learners work in four expert groups, each assigned one real-life Kenyan application of wave properties: (A) SONAR depth-sounding used by the Kenya Marine and Fisheries Research Institute (KMFRI) on Lake Victoria; (B) ultrasound scanning at Kenyatta National Hospital (KNH), Nairobi; (C) cats-eye road reflectors and speed cameras on the Nairobi–Mombasa highway; (D) seismic (earthquake) wave monitoring by the Kenya Meteorological Department (KMD). Groups research their application using available print or digital media, create a poster or digital slide, and present to the class (3 minutes presentation + 2 minutes peer questions). The teacher facilitates whole-class sensemaking discussion that maps each application back to reflection, refraction, or diffraction. The lesson closes with learners updating

the Driving Question Board with application cards and revising their cumulative wave model. Throughout, every application is explicitly reconnected to the fisherman's observations on Lake Victoria.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Phase 1 — Predict (Prior Knowledge Activation & Prediction)  VIDEO: Video: Bending Light Source: PhET Interactive Simulations (English)  Search ARES for similar videos  HTML: Seismic Waves (Flexbook) Source: CK-12  Search ARES for similar readings	Learners individually read a short stimulus card (1 paragraph, displayed on the board or printed) describing three puzzling Kenyan scenarios: (i) A KMFRI research vessel on Lake Victoria sends a pulse of sound downward and detects its echo from the lakebed 0.12 s later — the crew calculates the depth. (ii) A doctor at KNH moves an ultrasound probe over a patient's abdomen and sees a clear image of an organ on the screen. (iii) A seismologist at KMD in Nairobi detects ground waves from a tremor whose epicentre is near Suswa volcano in the Rift Valley, even though the waves have travelled through layers of different rock. Learners write a 2-minute silent individual prediction response to: 'Which wave property — reflection, refraction, or diffraction — is doing the most important work in each scenario, and why do you think so?' They record predictions in their science journals.	BEFORE THE LESSON: Write the three scenarios on the board or prepare printed stimulus cards (one per learner or one per pair). Display the driving question prominently. SAY: 'Before we look at any new information, I want you to use what you already know from our experiments in Lessons 3, 4, and 5. Read the three scenarios on the board silently. Then, in your science journal, write which wave property — reflection, refraction, or diffraction — you think is working in each scenario, and give one reason for each choice. You have exactly 2 minutes. Begin.' [START TIMER — 2 minutes silent individual writing.] After 2 minutes, SAY: 'Put your pens down. Let's hear some predictions.' Use cold-call on exactly 4 learners (one per table group, rotating from the previous lesson's last cold-call). Ask each: 'Which property did you choose for scenario (i/ii/iii), and what was your reasoning?' [WAIT 10 seconds after each question before accepting answers.] Record key prediction words on the board in a three-column table labelled REFLECTION REFRACTION DIFFRACTION. Do NOT confirm or correct yet. SAY: 'Hold on to these predictions — by the end of today you will know whether you were right, and more importantly, you will be able to explain why using real Kenyan evidence.'	Prediction journaling with cold-call sharing. This activates prior knowledge from Lessons 3–5, creates cognitive dissonance (learners discover they are uncertain about which property applies in complex real-life contexts), and provides the teacher with formative data on misconceptions to address during the gallery-walk debrief. Writing before speaking prevents dominant voices from anchoring peers' thinking.	Observable indicator: Every learner has a written prediction entry in their journal for all three scenarios before the cold-call begins. Teacher scans the room during the 2-minute writing phase — anyone not writing is prompted quietly. During cold-call, listen for: correct property identification (target: ≥ 3 of 4 cold-called learners correctly identify reflection for SONAR/ultrasound); misconceptions (e.g., 'refraction' for SONAR) are recorded on the board as 'Questions to resolve' to revisit in Phase 3.

Phase 2 — Observe (Expert-Group Research & Poster/Slide Creation)  VIDEO: Video: Bending Light Source: PhET Interactive Simulations (English)  Search ARES for similar videos  HTML: Seismic Waves (Flexbook) Source: CK-12  Search ARES for similar readings	Learners move into four pre-assigned expert groups of 4–6 (Groups A, B, C, D). Each group receives a Research Brief card specifying their assigned application, three guiding research questions, and a list of approved resources (textbook pages, printed fact sheets prepared by teacher, or specific URLs if internet is available). The four applications are: (A) SONAR by KMFRI on Lake Victoria — link to reflection; (B) Ultrasound scanning at KNH — link to reflection; (C) Cats-eye reflectors and speed cameras on the Nairobi–Mombasa highway — link to reflection and refraction; (D) Seismic wave monitoring by KMD — link to refraction and diffraction. Groups have 20 minutes to research and create ONE A3 poster or one digital slide that must include: (1) a diagram of how the waves travel in their application, labelling the relevant wave property; (2) a two-sentence explanation of HOW the wave property is being used; (3) a Kenyan context detail (name of institution, location, or specific benefit to a Kenyan community); (4) a one-sentence connection back to the Lake Victoria fisherman phenomenon. Groups assign roles: Researcher, Diagram-maker, Writer, Timekeeper/Presenter.	SAY: 'You are now expert groups. Each group will become the class authority on one application of waves in Kenya. Your job is to research, understand deeply, and then teach your classmates.' Distribute Research Brief cards and materials. SAY: 'Read your brief card carefully as a group. You have 20 minutes to research and create your poster or slide. Your poster must answer the three questions on your card, include a diagram, and — this is important — it must include one sentence that connects your application back to our fisherman at Lake Victoria. Go.' [START TIMER — 20 minutes.] Circulate every 3–4 minutes. At each group, ask ONE probing question without giving answers: Group A — 'How long did the sound pulse travel in total before the echo returned to the ship? What does that tell you about which wave property is at work?' Group B — 'Why does the ultrasound image change when the probe moves to a different angle? Which property causes that?' Group C — 'Why does the cat's eye glow at night even though it has no battery? Which property is responsible?' Group D — 'Seismic waves travel faster through dense rock than through soft sediment. What happens at the boundary — which property is that?' [WAIT 12 seconds after each question.] At the 15-minute mark, give a time warning: 'Five minutes remaining — make sure your Lake Victoria connection sentence is on the poster.' At 20 minutes, call time.	Jigsaw expert-group research. By making each group the exclusive 'expert' on one application, this strategy creates genuine interdependence during the gallery presentations — learners must pay close attention to peers because they will not have researched the other three applications themselves. The Research Brief card scaffolds the inquiry without removing cognitive challenge. Requiring a Lake Victoria connection sentence ensures every group explicitly links applied technology back to the anchoring phenomenon, reinforcing the storyline thread.	Observable indicators: (1) Every poster/slide contains all four required elements (diagram, property label, Kenyan context, phenomenon connection). Teacher checks this during circulation — any group missing an element is prompted: 'What wave property label is missing from your diagram?' (2) During probing questions, listen for correct use of vocabulary: 'reflection', 'refraction', 'angle of incidence', 'echo'. A group that cannot answer the probing question is directed: 'Look back at your experiment notes from Lesson 3 — what did the sound do when it hit the wall?' (3) Roles are being enacted — Timekeeper is tracking time, Diagram-maker is drawing. If one learner is dominating, intervene: 'I'd like to hear [name]'s idea about the diagram.'
Phase 3 — Explain	Each expert group presents their poster or slide to the class (3	SAY: 'We will now hear from each expert group. Audience: while you	Structured academic controversy presentation + T-chart whole-class	Observable indicators: (1) All Peer Feedback Cards are completed for

(Group Presentations & Whole-Class Sensemaking Discussion)  VIDEO: Video: Bending Light Source: PhET Interactive Simulations (English)  Search ARES for similar videos  HTML: Seismic Waves (Flexbook) Source: CK-12  Search ARES for similar readings	minutes), followed by 2 minutes of structured peer questions. Presenters stand beside their poster. The audience uses a Peer Feedback Card (prepared by teacher) to record: (a) which wave property the group described; (b) one thing they learned; (c) one question they want to ask. After all four presentations, the teacher facilitates a 10-minute whole-class sensemaking discussion that explicitly maps each application back to the wave-property headers (REFLECTION, REFRACTION, DIFFRACTION) and back to the Lake Victoria phenomenon. The teacher then revisits the predictions from Phase 1 and asks learners to confirm, correct, or nuance their original predictions in their journals.	listen, fill in your Peer Feedback Card — write the wave property, one thing you learned, and one question. Presenters: you have exactly 3 minutes, then 2 minutes for questions from the audience. I will time you.' [Run all four presentations with strict timing — use a visible timer projected or on the board.] After each presentation, select 2 peer questions from audience Peer Feedback Cards (cold-call on learners who have not yet spoken today — maintain a tally). After all four presentations (approx. 20 minutes total), SAY: 'Now let's make sense of what we've just heard together.' Draw a large T-chart on the board with columns: APPLICATION WAVE PROPERTY CONNECTION TO LAKE VICTORIA. Cold-call one learner per application row to complete the table (4 cold-calls, WAIT 12 seconds each). After the table is complete, SAY: 'Look at the first column. SONAR on Lake Victoria and ultrasound at KNH — what do they have in common with what our fisherman heard?' [WAIT 15 seconds — allow think time.] Cold-call 2 learners. Target answer: both use reflection of waves. SAY: 'And the seismic waves monitored by KMD bending through rock layers — what is the fisherman's water wave doing as it crosses from deep lake into the shallow bay?' [WAIT 12 seconds.] Target: both involve refraction. SAY: 'Now return to your journal prediction from Phase 1. Write one sentence: was your prediction correct, partially correct, or incorrect, and why? You have 90 seconds.' [TIMER 90 seconds.] Cold-call 2 learners to share their	mapping. The Peer Feedback Card turns passive listening into active evidence-gathering, mimicking the scientific practice of evaluating explanations (NGSS SEP 6: Constructing Explanations). The T-chart mapping session is a 'press for connections' strategy — the teacher deliberately asks learners to name the shared mechanism across different applications, building the generalisation that the same wave properties operate at all scales (microscopic ultrasound, geological seismic waves, everyday road reflectors). The journal self-correction closes the predict-observe-explain loop from Phase 1.	all four presentations (teacher collects at end of lesson — spot-check 8 cards). (2) T-chart on board is correctly populated — if a learner gives an incorrect property for a cell, do NOT correct immediately; instead ask the class: 'Does anyone want to add to or challenge that answer?' Only provide the correct answer if no learner does so within 10 seconds. (3) Journal self-correction: circulate during the 90-second writing window — target indicator is that ≥80% of learners have written a substantive sentence (not just 'yes' or 'no'). Flag learners who write nothing for follow-up in Phase 5.
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		self-corrections.		
Phase 4 — Driving Question Board (DQB) Update  VIDEO: Video: Bending Light Source: PhET Interactive Simulations (English)  Search ARES for similar videos  HTML: Seismic Waves (Flexbook) Source: CK-12  Search ARES for similar readings	Each expert group receives four blank index cards (one per application). They write the application name on the front of one card and the wave property it uses on the back, then walk to the class DQB and place each application card in the correct column beneath the wave-property header that was established in earlier lessons (REFLECTION, REFRACTION, DIFFRACTION). If an application uses more than one property (e.g., seismic waves use both refraction and diffraction), the group places a card in each relevant column with a brief note explaining the distinction. Learners also review any unanswered 'wonder' questions on the DQB from previous lessons and use a sticky dot to mark any that today's lesson has now answered. New questions that arose during today's presentations can be added as new sticky notes.	SAY: 'It's time to update our Driving Question Board. Each group: write your application name on one card and the property on the back. Walk to the board and place it where it belongs — under REFLECTION, REFRACTION, or DIFFRACTION. If your application belongs in more than one column, make a card for each column and add a short note explaining why.' [Allow 4 minutes for groups to place cards.] Once all cards are placed, stand at the board and narrate briefly: 'Look at what we now have under REFLECTION — SONAR, ultrasound, cats-eyes, and speed cameras. Under REFRACTION — seismic waves through rock layers, water waves at the bay edge, and cats-eyes again. Under DIFFRACTION — seismic waves around mountain masses, sound around the headland.' SAY: 'Now look at the wonder questions from Lesson 1. Use a sticky dot to mark any question that today's lesson has helped you answer. Add any new questions as sticky notes.' [Allow 3 minutes.] SAY: 'Our board is showing us a pattern — can anyone read that pattern back to me in one sentence?' [WAIT 15 seconds.] Cold-call 2 learners. Target: 'Wave properties are used in technology all around us in Kenya, and they are the same properties that explain what the fisherman saw at Lake Victoria.'	Physical DQB card placement as a knowledge-mapping ritual. The act of physically walking to the board and placing a card is a public knowledge claim — learners must commit to a category, which reveals understanding or misconceptions. The sticky-dot activity on prior wonder questions transforms the DQB from a static display into a dynamic record of growing understanding, making the storyline's progress visible to the whole class. This enacts NGSS SEP 8: Obtaining, Evaluating, and Communicating Information.	Observable indicators: (1) All four application cards are correctly placed — if a group places 'SONAR' under DIFFRACTION, ask publicly: 'Walk us through your reasoning — what does SONAR actually do with the sound pulse?' to elicit self-correction. (2) At least 3 prior 'wonder' questions receive sticky dots, indicating learners perceive their understanding as having grown. (3) The 'pattern' sentence from the cold-call correctly identifies the generalisation — if it does not, the teacher writes the correct generalisation on the board as a class consensus statement.
Phase 5 — Model Building (Cumulative Model)	Learners individually retrieve their cumulative wave model from their science journals (built incrementally across Lessons 1–5). They add a new layer to the	SAY: 'Open your science journals to your cumulative wave model. You have been building this model since Lesson 1. Today you are going to add an Applications Ring	Incremental model revision with an Applications Ring. Asking learners to add a ring to an existing model (rather than draw a new one) makes the cumulative nature of	Observable indicators: (1) All four applications appear in the Applications Ring with correct property arrows (teacher checks during circulation — note any

Revision)  VIDEO: Video: Bending Light Source: PhET Interactive Simulations (English)  Search ARES for similar videos  HTML: Seismic Waves (Flexbook) Source: CK-12  Search ARES for similar readings	model specifically for Lesson 6: an 'Applications Ring' — a ring of labelled icons or annotations surrounding their core wave diagram. Each application (SONAR/KMFRI, ultrasound/KNH, cats-eyes/Nairobi–Mombasa highway, seismic waves/KMD) is placed in the ring, connected by an arrow to the wave property it uses at the centre of the diagram. Learners also add one new annotation to the original Lake Victoria fisherman scene in their model, explaining which wave property explains the specific observation (e.g., an arrow from 'sound bending around the headland' to 'DIFFRACTION' labelled 'same physics as seismic waves around Rift Valley mountains'). Learners who finish early add a speculative annotation: 'One more Kenyan application I think uses [property] because...'	around the outside of your existing diagram. Draw a large circle around your current model. In that ring, place four icons or labels — one for each application we heard about today. Draw an arrow from each application to the wave property it uses in the centre of your diagram. Then add one new annotation to the fisherman scene — connect one of his observations to a real Kenyan technology that uses the same property. You have 8 minutes.' [START TIMER — 8 minutes.] Circulate and make targeted interventions: (1) To learners drawing the Applications Ring correctly but not annotating the fisherman scene: 'Good ring — now find the sound-bending-around-the-headland part of your original diagram and draw an arrow. What technology did we discuss today that uses that same behaviour?' [WAIT 10 seconds.] (2) To learners who have not started: 'Start with SONAR — draw a ship on Lake Victoria and show the sound pulse going down and the echo coming back. Which arrow in your centre diagram does that match?' At 7 minutes, SAY: 'One more minute — if you have finished, add a speculative annotation for a fifth Kenyan application not mentioned today.' At 8 minutes, cold-call 3 learners to hold up or describe their model additions. Close with: 'Your model now shows that the wave properties we used to explain the fisherman's puzzling observations are the same properties working inside medical machines, research vessels, road safety systems, and earthquake detectors across Kenya. Next lesson we will ask: what happens when waves affect	the storyline visible and concrete — learners literally see their understanding expanding outward from the core phenomenon. Requiring an arrow from each application to a central wave property enforces the mechanism-level understanding (not just 'SONAR uses sound') that is the target of the lesson. This enacts NGSS SEP 2: Developing and Using Models. The speculative annotation for early finishers extends thinking toward Lesson 7's environmental-impact focus.	incorrect arrows for remediation at the start of Lesson 7). (2) At least one new annotation is added to the original fisherman scene. (3) During the 3 cold-calls at the end, learners can describe their model addition in one sentence using the name of the wave property — not just the application name. Exit-check: as learners close journals, ask them to place a finger on the wave property they feel least confident about after today. Teacher notes the most common finger-placement (likely diffraction) to prioritise in the Lesson 7 warm-up.
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		the environment in ways that are harmful — and what can Kenya do about it?		
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your teaching journal:

1. **PRESENTATION QUALITY:** Were all four group presentations clear enough for the audience to correctly populate the T-chart? If Group D (seismic waves) struggled to connect KMD monitoring to the phenomenon, consider preparing a short 2-minute teacher-led bridge at the start of Lesson 7 that revisits refraction of seismic waves through rock layers alongside the water-wave refraction at the lake bay edge.
2. **COLD-CALL EQUITY:** Did you successfully cold-call learners who had not yet spoken? Review your tally — if the same 4–5 learners dominated responses, adjust the cold-call rotation strategy in Lesson 7 by pre-assigning a 'reporter' role within each group who must be the one to answer the first question directed at their group.
3. **DQB MISCONCEPTIONS:** Were any application cards placed in the wrong wave-property column? Note these for targeted questioning in Lesson 7's warm-up. The most likely error is placing SONAR under DIFFRACTION (because 'sound spreads from the boat') — if this occurred, prepare a brief diagram showing the difference between a directed pulse (reflection) and a wave spreading around a barrier (diffraction).
4. **MODEL DEPTH:** Do the cumulative model revisions show mechanism-level understanding (arrows from application → specific wave property → explanation of how it works) or surface-level labelling (just writing the application name)? If models are mostly surface-level, build in a structured 'model critique' at the start of Lesson 8 where learners swap journals and ask each other one 'how does it work?' question.
5. **PHENOMENON CONNECTION:** Did the Lake Victoria fisherman remain visible throughout the lesson, or did the applications feel disconnected from the original story? If the phenomenon felt distant, reopen Lesson 7 by re-reading the original phenomenon description and asking: 'Now that you know about SONAR, seismic monitoring, and ultrasound — which of the fisherman's three puzzling observations can you now fully explain using scientific language?'

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	In their expert groups, learners researched and presented one of four real Kenyan wave applications: KMFRI SONAR depth-sounding on Lake Victoria; ultrasound scanning at Kenyatta National Hospital; cats-eye reflectors and speed cameras on the Nairobi–Mombasa highway; and seismic wave monitoring by the Kenya Meteorological Department. They created posters or digital slides including wave diagrams, property labels, Kenyan context details, and a connection sentence to the Lake Victoria fisherman phenomenon, then presented to the class and responded to peer questions.
What did I learn?	Each real-life Kenyan application deliberately exploits one or more wave properties. SONAR and ultrasound use REFLECTION — a directed pulse is sent out and the returning echo carries information about distance or density. Cats-eye reflectors and speed cameras use REFLECTION of light waves from a curved surface back toward its source. Seismic wave monitoring uses REFRACTION (waves bend as they pass through rock layers of different density) and DIFFRACTION (waves spread around geological structures) — the same two properties that explain why the fisherman's water waves entered the sheltered cove and why sound bent around the rocky headland. All applications are grounded in the same wave equation and the same three properties investigated in Lessons 3–5.
How does this explain the phenomenon?	Wave properties are not abstract laboratory phenomena — they are the physical mechanisms inside medical imaging machines, marine research vessels, road safety infrastructure, and earthquake detection systems across Kenya. The fisherman on Lake Victoria was not experiencing something unusual; he was experiencing the same reflection, refraction, and diffraction that Kenyan engineers and scientists harness every day. Understanding these properties at a mechanistic level — knowing WHY a wave bends, bounces, or spreads — is what allows technology to be designed, improved, and used safely for the benefit of Kenyan communities.

LESSON 7 (80 minutes): Effects of Waves on the Environment & Mitigation

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate the real-world environmental impacts of wave energy — from coastal erosion at Mombasa and Lamu, to seismic wave damage in Nairobi, noise pollution in urban centres, and disruption of Lake Victoria's marine ecosystems — and evaluate evidence-based mitigation strategies, culminating in a structured debate about the environmental trade-offs of coastal wave-energy development in Kenya.
Knowledge	Learners will be able to: (a) explain the effects of waves on the environment including coastal erosion, seismic damage to infrastructure, noise pollution, and disruption of marine ecosystems; (b) describe mitigation measures such as sea walls, building codes, noise barriers, and marine protected areas; (c) evaluate the environmental trade-offs of wave-energy technology in a Kenyan coastal context.
Skills	Learners will be able to: (a) use print and non-print media to search for and synthesise information on the effects of waves on the environment; (b) construct and defend evidence-based arguments in a structured debate; (c) record and communicate research findings clearly; (d) update a Driving Question Board with new environmental-impact and mitigation evidence cards.
Attitudes	Learners will: (a) appreciate the responsibility of citizens and scientists to mitigate the harmful effects of wave energy on Kenyan communities and ecosystems; (b) demonstrate respect for peers' perspectives during debate; (c) show commitment to environmental stewardship in line with Kenya's national goals of promoting positive attitudes towards environmental protection.
Key Inquiry Question	1. How are waves applicable in our day-to-day life? 2. How do waves affect our environment?
Purpose in Storyline	In Lessons 1–6 learners built the wave equation, demonstrated reflection (echo method), investigated refraction and diffraction, and connected these properties to the Lake Victoria fisherman phenomenon. Lesson 7 pivots to consequences: now that learners can explain HOW waves behave, they interrogate WHAT waves do to the environment and communities over time — directly answering why the fisherman's storm matters beyond one evening on the lake, and why engineers in Mombasa, seismologists in Nairobi, and conservationists on Lake Victoria must all account for wave energy.
Safety Notes	No hazardous materials are used. Learners accessing non-print media must observe cyber-safety guidelines: use school-approved devices, do not share personal information online, and report any inappropriate content to the teacher immediately. During debate, enforce respectful-dialogue norms — no personal attacks; arguments must be evidence-based.

B. LESSON OVERVIEW

In this 80-minute lesson learners work in small groups to research the environmental effects of waves using print and non-print media, then engage in a structured three-team debate on Kenya's potential coastal wave-energy power station. The lesson is grounded throughout in the anchoring phenomenon: the Lake Victoria fisherman's storm night now serves as a gateway to understanding how the same wave behaviours — diffraction around headlands, refraction across depth gradients, sustained energy transfer — cause coastline erosion at Mombasa and Lamu, structural damage from seismic waves in Nairobi, noise-pollution harm in Kenyan cities, and ecosystem disruption in Lake Victoria. Learners update the Driving Question Board and begin drafting their cumulative explanatory model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Phase 1 — Predict (10 minutes)  VIDEO: Specific heat of water Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Seismic Waves (Flexbook) Source: CK-12  Search ARES for similar readings	Learners are shown a split display: (Left) a photograph of eroded coral-stone walls at Fort Jesus, Mombasa, with waves crashing at the base; (Right) the fisherman silhouette from the anchoring phenomenon card used in Lesson 1. Working individually for 3 minutes, learners write a silent prediction in their science journals responding to the prompt: 'The fisherman watched waves spread around the rocky headland all night. Over 50 years, what do you predict those waves have done to the headland, to the shoreline, to the fish habitat, and to the village behind the cliff? Give at least THREE specific predictions.' Learners then share predictions in a Think-Pair-Share with their table partner (2 minutes). Three pairs are cold-called to share with the class (1 minute each, teacher records key prediction words on the board under the headings: EROSION / ECOSYSTEMS / INFRASTRUCTURE / NOISE).	Display the split image before learners enter (pre-loaded on ARES platform or printed A3 poster). Read the prediction prompt aloud and say: 'Use what you already know about wave energy, diffraction, and refraction from our last six lessons — connect it to what waves DO over time.' Allow 3 minutes of silent individual writing — do NOT answer questions during this time; redirect with 'Trust your thinking.' After 3 minutes say: 'Turn to your partner. You have 2 minutes — share your three predictions. Listen carefully; you may need to defend them later.' Cold-call three pairs using the class register (row 1 seat 3, row 3 seat 1, row 4 seat 2 — vary each lesson). For each pair: 'Tell us your most surprising prediction and the wave property that drives it.' WAIT TIME: 12 seconds after each cold-call before accepting an answer. Record exact student words verbatim on the board — do not correct yet. Close with: 'Hold these predictions. By the end of today you will know which ones the evidence supports.'	Silent Prediction Journal + Think-Pair-Share. This activates prior knowledge of wave properties (Lessons 1–6) and creates a personal stake in the evidence-gathering phase. Recording predictions publicly on the board makes thinking visible and gives learners a concrete reference point to confirm, revise, or challenge during the Explain phase — a core epistemic practice of science.	Observable indicator: Every learner has at least three written predictions in their journal before the pair-share begins. Teacher scans journals during the pair-share circulations (look for: Does the prediction name a specific wave property — reflection, refraction, diffraction, or energy transfer? Does it reference a Kenyan context?). Cold-call responses reveal whether learners can causally link wave behaviour to environmental outcome. Flag any learner who cannot produce a causal link — pair them with a strong partner during the research phase.
Phase 2 — Observe / Research (25 minutes)  VIDEO: Specific heat of water Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Seismic Waves	Learners work in four home groups of approximately 4–5, each assigned ONE research focus card (pre-cut and laminated, or displayed on ARES): GROUP A — Coastal Erosion (Mombasa Old Town seawall, Lamu Island mangrove loss); GROUP B — Seismic Wave Damage (2006 Mozambique earthquake tremors felt in Nairobi, building code implications); GROUP C — Noise Pollution (wave-generated low-frequency noise near Kisumu port,	Distribute research focus cards and capture sheets before the phase begins. Say: 'Your job is not to copy — your job is to find EVIDENCE. Every piece of evidence must name the wave property that caused the environmental effect. Keep asking: could the fisherman on Lake Victoria see or hear evidence of this same process?' Circulate every 3–4 minutes, visiting each group at least twice. Use targeted prompts: to GROUP A — 'What	Jigsaw Research with Structured Capture Sheet. Each group becomes the class expert on one environmental domain. The four-column capture sheet enforces the disciplinary practice of citing evidence and linking mechanism (wave property) to effect — mirroring how Kenyan environmental scientists at NEMA or KMA produce impact assessments. The explicit 'Link to Fisherman Phenomenon' column keeps every research thread	Observable indicator: Each group's capture sheet has at least 3 rows of evidence, each row names a specific wave property (not just 'waves'). Teacher checks during circulation: Is the source column filled? Is the wave-property column specific (e.g., 'diffraction', 'amplitude', 'frequency', not just 'wave energy')? Groups that cannot fill the wave-property column are redirected to their ARES resource card glossary. Listen for misconceptions: 'erosion

(Flexbook) Source: CK-12 Search ARES for similar readings	urban traffic-wave noise barriers on Thika Superhighway); GROUP D — Marine Ecosystem Disruption (Lake Victoria wave disturbance of shallow spawning grounds, boat-wake waves displacing tilapia and Nile perch habitats). Each group uses available print media (pre-selected newspaper clippings from Daily Nation and Standard, NEMA environmental reports, KWS marine briefs) and/or non-print media (ARES offline resource cards with embedded images, infographics, and short text passages). Groups complete a RESEARCH CAPTURE SHEET with four columns: EVIDENCE FOUND SOURCE WAVE PROPERTY RESPONSIBLE LINK TO FISHERMAN PHENOMENON. Groups have 18 minutes for research, then 7 minutes to consolidate findings into a 90-second group report (4 bullet points maximum).	type of wave carries the erosion energy — transverse or longitudinal? How does diffraction around the Mombasa headland concentrate wave energy on the seawall?'; to GROUP B — 'Seismic waves are mechanical waves — how does the wave equation connect earthquake frequency to building damage?'; to GROUP C — 'Noise is a sound wave. What mitigation strategy uses the principle of wave absorption or reflection?'; to GROUP D — 'How does wave refraction in shallow Lake Victoria water change the energy hitting spawning grounds?' At 15 minutes, give a 3-minute warning: 'Wrap up research, begin your four bullet points.' At 18 minutes: 'Pens down on research — 7 minutes to prepare your 90-second group report.' WAIT TIME: when a group is stuck, ask the prompt question, then wait 15 seconds in silence before offering a scaffold. Do not give answers — redirect to the resource cards.	anchored to the driving question.	is caused by water, not waves' — address immediately with: 'What makes the water move in that pattern? What carries the energy to the seawall?'
Phase 3 — Explain (20 minutes)  VIDEO: Specific heat of water Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Seismic Waves (Flexbook) Source: CK-12 Search ARES for similar readings	Part A — Gallery Report (8 minutes): Each home group delivers their 90-second report to the class. The remaining groups listen and add any new evidence to their own capture sheets under a 'Peers Added' row. After all four reports, learners individually write a 2-sentence synthesis in their journals: 'The environmental effect I find most serious for Kenya is ____ because the wave property ____ causes ____ and the mitigation measure that best addresses it is ____.' Part B — Structured Debate (12 minutes): The class is reorganised into three debate teams of approximately 6–7:	For Gallery Reports: stand at the side, not the front — let groups present. After each report ask the class one cold-call question: 'Group [X] mentioned [wave property] — [student name], can you explain in your own words why that property specifically causes that environmental damage?' WAIT TIME: 12 seconds. Do not evaluate answers as 'right' or 'wrong' — instead respond with 'What evidence supports that?' or 'Does anyone want to add to or challenge that?' For the Structured Debate: assign teams deliberately — place confident speakers in the Adjudicator team so they model	Jigsaw Gallery Report + Structured Debate. The gallery report distributes expert knowledge across the class (Communication and Collaboration core competency). The structured debate operationalises Critical Thinking and Problem Solving — learners must weigh evidence, anticipate counter-arguments, and reason under uncertainty, mirroring the real decision-making process Kenya faces on renewable energy and coastal management. Explicitly naming this mirrors NGSS Science and Engineering Practice 7 (Engaging in Argument from Evidence).	Observable indicators during Gallery Reports: Does each group's opening sentence name the wave property AND the Kenyan location? During the debate: Does each team's opening statement include at least one piece of evidence from the research phase (not just opinion)? Does the Adjudicator team ask at least one question that references a specific mitigation measure? After the debate: Do learners' journal synthesis sentences correctly name both a wave property and a mitigation measure? Collect 6 journals (one from each home group) at the end

	TEAM PROPONENT ('Kenya SHOULD build a coastal wave-energy power station at Malindi — the benefits outweigh environmental costs'), TEAM OPPONENT ('Kenya should NOT build it — the environmental and community costs are too high'), TEAM ADJUDICATOR/QUESTIONER ('We evaluate the evidence and ask the hard questions'). Each team has 2 minutes to confer and prepare ONE opening statement (max 3 evidence-based points) and ONE anticipated counter-argument. Debate runs: Opening statements (1 minute per side) → Adjudicator questions (2 minutes) → Rebuttals (1 minute per side) → Adjudicator verdict with reasoning (1 minute). All learners record the adjudicator's final reasoning in their journals.	evidence-based reasoning. During the 2-minute preparation say: 'Your arguments MUST reference a specific wave property and a specific Kenyan location or data point — no generic statements accepted.' During the debate, enforce the time strictly (use a visible timer). After the Adjudicator verdict, ask the whole class: 'Does the verdict change any of your original predictions from Phase 1? Write YES or NO and one sentence in your journal — you have 60 seconds.' This closes the predict-explain loop. Close Phase 3 with: 'Science does not always give us a clean answer — engineers at Kenya Power, conservationists at KWS, and fishermen on Lake Victoria may all look at the same wave-energy data and reach different conclusions. That tension is exactly what our DQB should capture.'		of this phase for rapid teacher review before the next lesson.
Phase 4 — Driving Question Board Update (10 minutes)  VIDEO: Specific heat of water Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Seismic Waves (Flexbook) Source: CK-12  Search ARES for similar readings	Learners return to their home groups. Each learner writes TWO new DQB cards on sticky notes or index cards provided: CARD TYPE 1 (orange) — an ENVIRONMENTAL IMPACT card stating one effect of waves on the Kenyan environment discovered today, in the format: '[Wave property] causes [environmental effect] at [Kenyan location] because [mechanism].' CARD TYPE 2 (green) — a MITIGATION card stating one human response, in the format: '[Mitigation measure] reduces [wave effect] by [how it works physically].' Groups share cards within the group (1 minute), select the two strongest cards per group (one orange, one green), and one representative from each	Distribute orange and green cards (or direct learners to write on sticky notes in two colours). Say: 'Every card must have a Kenyan place name and a wave property — if it doesn't, rewrite it before it goes on the board.' As groups write (3 minutes), circulate and check: Is the mechanism column filled? Are learners confusing 'cause' with 'effect'? Prompt: 'You've written that erosion damages the seawall — but what is the wave DOING physically to cause erosion? Is it the amplitude? The frequency? The diffraction pattern?' When groups post cards, cold-call two students (not the representatives) to explain a card posted by a different group: 'Amina, can you explain in your own words what	Structured DQB Card Protocol (Colour-Coded by Function). Colour-coding impact vs. mitigation cards makes the causal structure of the evidence visible at a glance and supports the distinction between scientific explanation (how waves cause harm) and engineering response (how humans mitigate harm) — directly aligning with NGSS ETS1 engineering design thinking. The public posting ritual reinforces collective knowledge-building: the DQB belongs to the class, not to any individual.	Observable indicator: Every learner has produced two written cards before the group selection process. Teacher checks that orange cards name a wave property (not just 'waves'), a Kenyan location, and a mechanism. Green cards must name a physical principle (e.g., 'a sea wall reflects incoming wave energy before it reaches the shoreline' — not just 'sea walls protect the coast'). Cold-call responses reveal whether learners can explain a peer's card — if a learner cannot paraphrase a card, the card's author is asked to rewrite it in simpler language.

	group walks to the DQB and places their cards in the correct section, reading each card aloud in one sentence. The class then reviews the DQB as a whole for 2 minutes: teacher facilitates with 'Which new card most directly answers our driving question about the Rift Valley mobile-phone signal fading behind a hill? Which card connects back to the fisherman's storm night?'	Group B's mitigation card means and why it works?' WAIT TIME: 12 seconds. After the DQB review, point to the fisherman phenomenon card still anchored at the top of the DQB and say: 'The fisherman saw one night of wave energy. These cards represent what decades of that same energy do. Has our model of waves gotten bigger, or more complicated, or both?' Accept 2–3 brief verbal responses.		
Phase 5 — Model Building (15 minutes)  VIDEO: Specific heat of water Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Seismic Waves (Flexbook) Source: CK-12  Search ARES for similar readings	Learners open their cumulative model notebooks (begun in Lesson 1 and revised each lesson). They have 10 minutes to complete a STRUCTURED MODEL DRAFT with three required zones drawn on a new page: ZONE 1 — 'The Wave Source & Propagation' (carries forward models from Lessons 1–6: wavefronts, wavelength, frequency, amplitude, reflection, refraction, diffraction diagrams). ZONE 2 — 'Environmental Impacts' (NEW today): learners draw at least TWO annotated arrows from a wave source to an environmental target (e.g., storm waves → diffraction around headland → concentrated energy on Lamu shoreline → erosion of mangrove roots). Each arrow must be labelled with the responsible wave property. ZONE 3 — 'Human Mitigation Layer' (NEW today): learners add at least ONE engineering or policy intervention to their diagram — drawn as a barrier, label, or annotation — and write one sentence explaining the physical principle by which it works. After 10 minutes, learners do a 3-2-1 Model Peer Review with their table partner: identify 3	Before the phase begins, display on the board (or ARES screen) a skeleton example of the three-zone model layout — show only the zone labels and a blank arrow, NOT a completed model. Say: 'Your model is YOUR scientific explanation — it should reflect your thinking today, not a copy of anything on the board.' During the 10-minute drawing time, circulate with the following specific checkpoints: (1) Is ZONE 1 connected to the phenomenon — does it show wave propagation from a storm source? (2) Does ZONE 2 show at least two environmental targets with named wave properties? (3) Does ZONE 3 show a mitigation intervention with a physical explanation? Prompt struggling learners: 'Where does the wave energy go when it hits the sea wall? Draw that.' Prompt advanced learners: 'Can your model also show why mitigation sometimes fails — for example, when wave energy is too high for the sea wall to absorb?' During the peer review (3 minutes), circulate and listen for the '1 question still unanswered.' Cold-call two learners to share their unanswered question with the	Three-Zone Cumulative Model Revision + 3-2-1 Peer Review. The three-zone structure scaffolds the transition from descriptive (what waves do) to explanatory (why, via named properties) to applied (how humans respond) — the full explanatory arc of the sub-strand. The 3-2-1 peer review builds metacognitive awareness (Learning to Learn core competency) and produces a class inventory of unresolved questions that feed directly into Lesson 8's DQB. Explicitly naming what the model can and cannot yet explain is a hallmark of scientific modelling practice (NGSS SEP 2: Developing and Using Models).	Observable indicators: (1) Zone 2 contains at least 2 annotated arrows with named wave properties — not generic 'wave damage'. (2) Zone 3 contains at least 1 mitigation intervention with a written physical explanation. (3) The '1 question' from the peer review is recorded in the journal — this is the primary exit-ticket for Lesson 7. Teacher collects the '1 unanswered question' from every learner (read from journals during the peer-review circulation or collected on a slip) to inform the Lesson 8 opening. Any learner whose Zone 2 contains no named wave property is flagged for targeted support at the start of Lesson 8.

	things the model shows correctly, 2 things that could be made clearer, 1 question the model still cannot answer. Learners record the '1 question' in their journal as their personal contribution to next lesson's DQB.	class. Write those questions on the DQB under 'Still Wondering'. Close the lesson with: 'Next lesson we apply wave properties to technology — mobile phones, sonar on Lake Victoria, and medical imaging. Your models today gave you the environmental foundation. Bring your journals.' Issue the reflection prompt for homework: 'In 3–5 sentences, explain to the Kisumu fisherman why the same wave behaviour that made his storm night strange is also responsible for the erosion he might notice on the shoreline near his village — and what Kenya could do about it.'		
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record notes in your professional journal:

1. **EVIDENCE QUALITY** — Did learners' research capture sheets name specific wave properties (diffraction, refraction, amplitude, frequency) as causes of environmental effects, or did they remain at a generic 'waves cause damage' level? Which groups needed the most redirection, and what resource gap drove that?
2. **DEBATE RIGOUR** — Did the structured debate produce evidence-based arguments, or did teams rely on opinion? If arguments were weak, consider providing a 'claim–evidence–reasoning' sentence frame in Lesson 8's debate scaffold.
3. **DQB CARDS** — Review the orange and green cards posted on the DQB. How many correctly named both a wave property AND a Kenyan location? Cards that did not meet this standard should be returned to groups for revision at the start of Lesson 8.
4. **MODEL PROGRESSION** — Compare learners' Zone 2 and Zone 3 additions to their Lesson 6 models. Is there evidence of cumulative understanding — are learners integrating earlier vocabulary (wavelength, frequency, diffraction) into their environmental-impact diagrams — or are the two zones disconnected from earlier model work?
5. **PHENOMENON CONNECTION** — Did the fisherman phenomenon remain live throughout the lesson, or did it fade once learners moved into research? If it faded, plan a 90-second phenomenon reconnect at the start of Lesson 8 before moving to applications.
6. **HOMEWORK QUALITY** (review before Lesson 8) — The fisherman explanation homework reveals whether learners can synthesise wave behaviour, environmental impact, and mitigation into a coherent causal narrative. Use strong examples as model texts at the start of Lesson 8.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners researched print and non-print sources to find evidence of wave impacts on the Kenyan environment: coastal erosion at Mombasa and Lamu driven by wave diffraction and high-amplitude storm waves; seismic wave damage to Nairobi infrastructure linked to wave frequency and building resonance; noise pollution from low-frequency sound waves near Kisumu port and along the Thika Superhighway; and disruption of Lake Victoria's tilapia and Nile perch spawning grounds caused by wave refraction in shallow water and boat-wake wave energy.
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What did I learn?	Each environmental impact is caused by a specific, nameable wave property: diffraction concentrates wave energy on unprotected shorelines; high-amplitude waves carry sufficient energy to erode coral-stone foundations; seismic wave frequency can match structural resonance frequencies in buildings; noise is a longitudinal wave whose energy disrupts biological systems. Mitigation measures work by applying the same wave physics in reverse — sea walls reflect or absorb wave energy before it reaches the shoreline; building codes specify materials whose resonance frequencies differ from local seismic wave frequencies; noise barriers reflect and absorb sound waves; marine protected areas reduce boat-wake wave disturbance of shallow spawning grounds.
How does this explain the phenomenon?	Learners used the wave equation and wave-property vocabulary from Lessons 1–6 to construct causal arguments in a structured debate about the environmental trade-offs of a Kenyan coastal wave-energy power station. The debate required learners to weigh the same wave energy that causes coastal erosion as a potential renewable electricity source — illustrating that wave energy is neither inherently harmful nor beneficial, but that human decisions about technology and mitigation determine outcomes. The fisherman phenomenon was extended: the same diffraction and refraction processes that made the storm night puzzling are the processes that, sustained over decades, reshape Kenya's coastlines, threaten Lamu's UNESCO heritage buildings, and alter the ecology of Lake Victoria. Learners updated the DQB with environmental-impact and mitigation cards and revised their cumulative models to include a human-mitigation layer, advancing the class's collective explanation of the driving question.

LESSON 8 (80 minutes): Model Building, Numerical Consolidation & DQB Closure — How Wave Science Keeps Kenyan Communities Safer

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To consolidate the full unit of wave science through individual numerical problem-solving, paired cumulative model building, DQB closure, and a final written Scientist's Statement that answers the anchoring phenomenon and connects wave behaviour to real Kenyan community applications.
Knowledge	Learners explain the terms used in waves, interpret the wave equation as used in science, and describe the effects of reflection, refraction, and diffraction of waves in real-life Kenyan contexts (KICD Sub-Strand 3.3a, b, d).
Skills	Learners calculate wave characteristics using the wave equation (including echo-distance and wavelength shift on refraction), demonstrate the effects of wave properties in annotated diagrams, and evaluate evidence gathered across the unit to construct a final explanatory model (KICD Sub-Strand 3.3b, c, d).
Attitudes	Learners appreciate the applications of waves in real life — including safe road use, coastal community protection, and mobile-phone infrastructure — and take responsibility for communicating science accurately to non-specialist audiences such as Kisumu fishermen (KICD Sub-Strand 3.3e; National Goal 8: positive attitudes towards environmental protection).
Key Inquiry Question	1. How are waves applicable in our day-to-day life? 2. How do waves affect our environment?
Purpose in Storyline	This is the final lesson of the unit. Learners have accumulated evidence across seven lessons about the Lake Victoria anchoring phenomenon — the fisherman who heard sound bend around a rocky headland, saw water waves refract into the shallow bay, and wondered why his mobile signal fades behind a Rift Valley hill. Today they synthesise ALL of that evidence: first individually (numerical problems under timed conditions), then in pairs (final annotated model), then as a class (DQB closure), and finally as individual scientists writing a plain-language explanation to the Kisumu fisherman. The lesson closes the driving question loop and celebrates the learners' journey from puzzlement to expertise.
Safety Notes	No hazardous materials used in this lesson. Learners work at desks and on paper. Remind learners to handle rulers and compasses (used for diagram arcs) carefully. Ensure adequate lighting for detailed diagram work. If any digital devices are used for the DQB display, handle cables safely and keep walkways clear.

B. LESSON OVERVIEW

Lesson 8 is the unit capstone. It opens with a brief 'flashback' to the Lake Victoria phenomenon before learners individually complete a timed mixed numerical problem set covering the wave equation ($v = f\lambda$), echo-distance calculation, and wavelength shift on refraction — all set in Kenyan contexts (Kisumu harbour sonar, Nairobi radio broadcast, Mombasa port depth-sounding). Learners then work in pairs to produce or substantially revise their cumulative annotated diagram, which must label reflection, refraction, and diffraction on the original Lake Victoria scene, embed the wave equation, and include at least two real-life applications and one environmental impact with a mitigation measure. The class reconvenes for Driving Question Board (DQB) Closure: the teacher reads every 'I wonder...' card posted in Lesson 1 aloud; cold-calls a minimum of six different students to answer using unit evidence; unanswered questions are flagged for further inquiry. The lesson ends with each learner writing a 'Scientist's Statement' — one paragraph in plain language addressed to the Kisumu fisherman — explaining how wave science (reflection, refraction, diffraction, the wave equation) keeps coastal communities safer. Teacher collects statements as a summative artefact.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Phase 1 — Predict: Flashback & Final Prediction (0–10 min)  VIDEO: Video: Bending Light Source: PhET Interactive Simulations (English)  Search ARES for similar videos  HTML: Landforms from Wave Erosion and Deposition (Flexbook) Source: CK-12  Search ARES for similar readings	Learners enter to find the anchoring phenomenon image projected on the board: the Kisumu fisherman on Lake Victoria during the storm, the rocky headland, the shallow bay, and the distant lightning flash. Beneath it the original driving question is displayed. Each learner opens their exercise book to a fresh page headed 'Lesson 8 — What I Now Know' and, working silently for 3 minutes, writes three sentences: (1) what they originally could NOT explain about the phenomenon (Lesson 1 prediction), (2) what they can NOW explain using scientific language, and (3) one thing they are still uncertain about. Learners do NOT share yet — this is a private self-audit. The teacher then invites two volunteers to read their Lesson 1 predictions aloud from their books (these were recorded in Lesson 1) so the class can hear how far they have come.	As learners enter, say: 'Take your seats. You have 3 minutes of silence. Look at the fisherman on the board. In your book, write what you could NOT explain in Lesson 1, what you CAN explain now, and what still puzzles you. Go.' Set a visible timer for 3 minutes. Circulate silently — do not prompt. After 3 minutes, say: 'Pens down. I need two brave volunteers — someone who remembers writing a prediction in Lesson 1 that was completely wrong. Who will share?' Wait 10 seconds. Cold-call if no hands: 'Amara, read us your Lesson 1 prediction word for word.' After each volunteer reads, say: 'Thank you. Hold that thought — by the end of today you will see exactly how that prediction evolved.' Do NOT correct or explain yet; preserve curiosity for the model phase.	Self-Audit / Reflective Journaling — learners compare their prior mental model (Lesson 1) with their current understanding, making metacognitive growth visible before the formal consolidation begins. This activates all unit knowledge as a network rather than isolated facts.	Observable indicator: each learner has three written sentences. Teacher scans books during circulation — flag any learner whose 'what I now know' column is sparse (fewer than one scientific term); seat them next to a stronger peer for the paired model phase. The two volunteer readings reveal the class's starting point and signal whether the unit narrative has landed.
Phase 2 — Observe (Individual Numerical Problem Set): Mixed Wave Calculations (10–30 min)  VIDEO: Video: Bending Light Source: PhET Interactive Simulations (English)  Search ARES for similar videos	The teacher distributes a one-page printed (or dictated) problem set of five questions. Learners work individually and silently for 18 minutes. All questions are set in Kenyan contexts and draw on knowledge from across the unit: Q1 (Wave equation — radio): A Nairobi FM station broadcasts at a frequency of 98.4 MHz. Given that radio waves travel at 3.0×10^8 m/s, calculate the wavelength of the broadcast signal. Show all working.	Distribute/dictate the problem set and say: 'You have 18 minutes. Work individually and silently. Show all working — answers without working earn zero marks. This is your personal evidence of what you have learned.' Set timer. Circulate every 2 minutes. If a learner is stuck on Q1 after 4 minutes, whisper the prompt: 'Write $v = f\lambda$. What do you know? What are you finding?' Do not give answers. After 18 minutes, say: 'Pens down. I will now reveal the mark scheme.' Display answers one by one on the board (pre-written on a flip-chart or projected).	Worked-Example Self-Marking with Error Analysis — learners do not simply receive answers; they identify the precise step where their reasoning diverged from the correct path. This is a well-evidenced strategy for closing procedural gaps and builds self-regulation (NGSS Science and Engineering Practice 5: Using Mathematics and Computational Thinking).	Observable indicators: (1) All five questions attempted — flag any learner with two or more blanks for targeted support during the model phase. (2) Q3b correct answer is $\lambda = 2.25$ m ($v=0.90/f=0.40$); Q4a = 85 m; Q2 = 107.1 m — teacher checks these key answers during circulation. (3) Self-marking colour change visible in books. Cold-call responses reveal whether the class can articulate the wave equation and the rule that frequency is unchanged on refraction.

 **HTML:**
[Landforms from Wave Erosion and Deposition \(Flexbook\)](#)

Source: CK-12
 [Search ARES for similar readings](#)

Q2 (Echo distance — sonar): A sonar device on a Kenya Ferry Services vessel in Mombasa harbour emits a sound pulse that returns as an echo after 0.28 s. Given that the speed of sound in seawater is 1 530 m/s, calculate the depth of the harbour floor below the vessel.

Q3 (Refraction — wavelength shift): A water wave in the deep open water of Lake Victoria has a frequency of 0.40 Hz and a wavelength of 3.75 m. As it crosses into the shallow bay near Kisumu, its speed decreases to 0.90 m/s. Calculate: (a) the speed of the wave in deep water, (b) the wavelength of the wave in the shallow bay. State whether frequency changes on refraction and explain why.

Q4 (Wave equation — applied): A student in Kericho claps once near a large concrete wall and hears the echo after 0.50 s. If the speed of sound in air is 340 m/s, calculate: (a) the distance from the student to the wall, (b) the minimum frequency a sound must have for a human ear to detect it as a distinct echo (given the ear needs at least 0.1 s between original sound and echo).

Q5 (Application + environment): A mobile-phone tower near a hill in the Rift Valley transmits microwaves of frequency 1.8 GHz. (a) Calculate the wavelength of these microwaves (speed of light = 3.0×10^8 m/s). (b) A village is in a valley directly behind the hill. Using the concept of diffraction,

For each answer, say: 'Mark it in red. If you got it right, put a tick. If wrong, write the correct answer and circle the step where you went wrong.' After self-marking, ask: 'Raise your hand if you scored 5 out of 5.' Pause. 'Four?' Pause. 'Three or below — do not worry; the model phase will help you consolidate.' Cold-call three learners: 'Kipkorir, walk us through your working for Q3b in one sentence.' Wait 10 seconds between each cold-call. Collect problem sets as formative evidence.

	explain in two sentences why some signal still reaches the village. (c) Name ONE environmental impact of increased microwave tower density near Rift Valley communities and suggest ONE mitigation measure. Learners check answers against a self-mark key that the teacher reveals on the board after 18 minutes. Learners mark their own work in a different colour and identify which question(s) they need to revisit.			
Phase 3 — Explain (Paired Cumulative Model Building & Final Annotation): The Complete Lake Victoria Diagram (30–55 min)  VIDEO: Video: Bending Light Source: PhET Interactive Simulations (English)  Search ARES for similar videos  HTML: Landforms from Wave Erosion and Deposition (Flexbook) Source: CK-12  Search ARES for similar readings	Learners pair with a neighbour. Each pair retrieves their cumulative annotated diagram begun in earlier lessons (or starts a fresh A3-size diagram if earlier versions are incomplete). The diagram must show the Lake Victoria anchoring phenomenon scene — the open lake, the rocky headland, the shallow bay, the cove, the Kisumu shoreline, and the distant storm — and must now include ALL of the following, each clearly labelled with a one-sentence scientific explanation: (A) REFLECTION: The sound of thunder reflecting off the rocky cliff face, forming an echo that the fisherman hears. Label the angle of incidence and angle of reflection. Write the echo equation: $d = (v \times t) / 2$. (B) REFRACTION: Water waves bending as they cross from deep open lake into the shallow bay. Draw the wavefronts slowing and bending toward the normal. Label: 'Speed decreases → wavelength	Say: 'You have 20 minutes. Work in pairs. Your diagram must have all six labelled elements — I will be checking each pair. If your diagram from earlier lessons is strong, refine and complete it. If it needs rebuilding, start fresh on the A3 sheet.' Distribute A3 paper if needed. Circulate methodically. At each pair, ask ONE of these probing questions (rotate to avoid repetition): 'Why does the wavelength change in the shallow bay but the frequency does not?' 'How does your diagram show that diffraction depends on the gap size relative to wavelength?' 'Which of your two applications is most directly relevant to a Kisumu fisherman's daily life — defend your choice.' Wait 10 seconds for each response before prompting. At 15 minutes, give a 5-minute warning: 'Five minutes — make sure all six elements are labelled and your wave equation box has actual numbers.' At 20 minutes: 'Pens down. Hold your diagrams up.' Conduct a rapid gallery-walk: learners stand, hold diagrams facing outward, and spend 60	Cumulative Annotated Model Revision (NGSS Science and Engineering Practice 2: Developing and Using Models) — the diagram is not a new task but a living model that has grown across the unit. Requiring all three wave properties on one Kenyan scene forces learners to see them as a unified framework rather than three separate topics. The gallery-walk adds peer validation and exposes learners to alternative representational choices without formal evaluation pressure.	Observable indicators: (1) All six elements present and labelled — teacher marks a checklist mentally for each pair during circulation. (2) Refraction diagram shows wavefronts bending toward the normal (not away) — a common error to watch for. (3) Diffraction diagram shows circular spreading past the headland, not straight-line rays. (4) Wave equation box contains numbers, not just symbols. (5) Environmental impact is specific to a Kenyan coastal or lake context, not generic. Flag any pair missing elements (C) or (F) — they will need to add these before submitting.

	decreases → direction changes → frequency stays the same.' Write $v_1/v_2 = \lambda_1/\lambda_2$. (C) DIFFRACTION: Sound waves spreading around the headland into the cove where no straight-line path exists. Draw circular wavefronts spreading past the gap. Write: 'Diffraction is greatest when the wavelength $\approx$ the gap size.' (D) WAVE EQUATION BOX: In a box somewhere on the diagram, write $v = f\lambda$ with a worked example using numbers from Q1 or Q3 of the problem set. (E) TWO REAL-LIFE APPLICATIONS: Each labelled with the wave property it uses. Examples: sonar depth-sounding (Mombasa port) uses reflection; radio signal bending around hills (Rift Valley) uses diffraction; mirages along Mombasa Road use refraction. (F) ONE ENVIRONMENTAL IMPACT + ONE MITIGATION MEASURE: e.g., 'Storm surges (wave energy) damage Kisumu fishing villages → mitigation: mangrove belt restoration along the shore acts as a wave-energy buffer.' Pairs have 20 minutes. Teacher circulates, spending at least 90 seconds with each pair, checking for the six required elements and asking probing questions.	seconds silently observing two neighbouring pairs' work. Then say: 'In one sentence, tell your partner the single strongest element you saw on another pair's diagram.' Cold-call three pairs for their observation.		
Phase 4 —	The teacher brings the class back	Say: 'We are now going to close	Driving Question Board Closure (a	Observable indicators: (1)

Driving Question Board (DQB) Closure: Answering Every 'I Wonder...' (55–70 min)  VIDEO: Video: Bending Light Source: PhET Interactive Simulations (English)  Search ARES for similar videos  HTML: Landforms from Wave Erosion and Deposition (Flexbook) Source: CK-12  Search ARES for similar readings	together and displays (or reads aloud from cards) every 'I wonder...' question posted on the DQB since Lesson 1. Questions were generated by learners during Lessons 1–7 and represent the full arc of the unit's inquiry. For each question, a student is cold-called to answer using evidence from the unit. Learners listen, note whether they agree or can add to each answer, and raise a hand to supplement. After each answered question the teacher sticks a green 'ANSWERED' sticker (or writes a tick) on the card. Any question for which no learner can provide a satisfactory evidence-based answer receives a yellow 'FOR FURTHER INQUIRY' sticker. At the end, learners silently count: How many questions answered? How many flagged? The teacher asks: 'What does it mean for science that some questions are still open?' Learners discuss briefly in pairs before one pair shares with the class.	our Driving Question Board. I will read each card. When I call your name, stand up and answer the question using evidence — not opinion, evidence. You may refer to your diagram or your problem-set answers.' Read the first DQB card aloud slowly. Wait 5 seconds. Cold-call: 'Zawadi, what is your answer and what is your evidence?' After Zawadi responds, say: 'Does anyone want to add a piece of evidence Zawadi did not mention?' Wait 10 seconds. Cold-call a second learner to add or agree. Stick the green sticker. Repeat for ALL DQB cards, ensuring a minimum of 6 DIFFERENT students are cold-called as primary responders (record names on a tally to avoid repetition). For any card that stumps the class, say: 'This is a great sign — science always has open questions. We flag this for further inquiry.' Stick the yellow sticker. After the last card, ask: 'Kamau, in one sentence: why does science always leave some questions open?' Wait 10 seconds. After response, say: 'Exactly. That is what makes you scientists, not just students.'	cornerstone of Phenomenon-Based Learning / Project Zero Thinking Routines) — publicly answering the questions learners themselves posed closes the inquiry loop, validates their curiosity as the engine of learning, and makes the unit's knowledge growth socially visible. Flagging unanswered questions models authentic scientific practice and intellectual humility (NGSS Science and Engineering Practice 8: Obtaining, Evaluating, and Communicating Information).	Minimum 6 different students provide primary cold-called responses — teacher tally. (2) Answers reference specific evidence: a named experiment, a calculation result, a diagram label — vague answers ('because of waves') are redirected: 'Give me the specific evidence.' (3) The ratio of green to yellow stickers reveals the class's collective coverage of the unit; more than 2 yellow stickers on core wave-property questions signals a gap that the teacher notes for follow-up. (4) The pair discussion on 'what does it mean that some questions are open' reveals epistemic sophistication — look for learners who reference the nature of science rather than expressing failure.
Phase 5 — Model Building (Final Written Output): The Scientist's Statement (70–80 min)  VIDEO: Video: Bending Light Source: PhET Interactive Simulations (English)	Working individually, each learner writes a 'Scientist's Statement' — a single paragraph of 80–120 words — addressed directly to the Kisumu fisherman from the anchoring phenomenon. The statement must: (1) explain in plain language why the thunder sound bent around the rocky headland (diffraction); (2) explain why the water waves changed direction at the edge of the shallow bay (refraction); (3) connect these ideas to why mobile-phone signals	Say: 'You have 10 minutes. You are now a scientist. Write directly to the Kisumu fisherman — the man from our very first lesson. He is intelligent but has not studied waves. Use plain language. Explain the bending of sound, the turning of water waves, the fading phone signal, and one way wave science protects communities like his. Your paragraph must have all four elements. Go.' Set timer for 8 minutes. At 8 minutes: 'Pens down. Swap with your neighbour.'	Scientist's Statement / Audience-Aware Science Writing (NGSS Science and Engineering Practice 8: Communicating Information) — requiring learners to translate formal scientific knowledge into plain language for a specific non-expert audience is one of the highest-order synthesis tasks in science education. The peer-tick formative check adds immediate self-correction without teacher bottleneck. The explicit connection back to the Lesson 1 phenomenon	Observable indicators: (1) All four required elements present (diffraction explanation, refraction explanation, mobile-phone connection, community-protection measure) — teacher reads and marks against a four-point rubric after collection. (2) Plain language used: no unexplained jargon — e.g., 'refraction' must be followed by a plain-language clause, not left as a standalone term. (3) Kenyan specificity: the statement references Kisumu, Lake Victoria,

 Search ARES for similar videos  HTML: Landforms from Wave Erosion and Deposition (Flexbook) Source: CK-12  Search ARES for similar readings	fade behind Rift Valley hills and what engineers do about it; (4) mention one specific way that understanding wave science helps protect coastal communities (e.g., mangrove belts, tsunami early-warning sonar, storm-surge barriers). The statement must NOT use jargon without explanation — it must be understandable to a fisherman with no formal science education. Learners then share their statement with the learner next to them, who reads it and places a tick next to each of the four required elements they can find. Learners hand the statements to the teacher as the final summative artefact of the unit.	Read their statement. Tick every one of the four required elements you can find. If an element is missing, put a question mark in the margin. You have 2 minutes.' At 10 minutes: 'Return the statement. Look at your ticks and question marks. If you have time, add one sentence to cover any missing element.' Collect all statements. As you collect, say to the class: 'You walked into Lesson 1 unable to explain why a fisherman on Lake Victoria could hear thunder from behind a cliff. You can now explain it — in scientific language AND in plain language. That is what it means to understand science. Well done.' End the lesson.	closes the narrative arc of the entire unit.	the Rift Valley, or another local context — generic statements ('waves help communities') score lower. (4) Peer ticks visible on the paper confirm the statement was readable to a peer audience.
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your journal:

1. **NUMERICAL CONSOLIDATION:** Which of the five problem-set questions had the highest error rate during self-marking? Was it Q3b (wavelength shift on refraction — the most conceptually demanding)? If so, plan a 10-minute remediation slot in the next available lesson where you work through that question step-by-step on the board using the Kisumu shallow-bay context.
2. **DQB CLOSURE:** How many of the DQB questions received green 'ANSWERED' stickers? If fewer than 70% of cards were answered satisfactorily, identify which wave property (reflection, refraction, or diffraction) was least well represented in student answers — this flags a gap in the unit's instructional sequence that you should address when you teach this sub-strand in the next academic year.
3. **COLD-CALL EQUITY:** Review your tally of cold-called students. Were the minimum 6 primary respondents drawn from across the gender, ability, and seating distribution of the class? If the same learners dominated responses, plan a random-selection strategy (e.g., name sticks) for future DQB closure sessions.
4. **SCIENTIST'S STATEMENT QUALITY:** After marking the statements, what proportion of learners achieved all four required elements? Did learners maintain plain language throughout, or did they default to jargon? Share two exemplary statements (anonymised) at the start of the next General Science lesson as models of scientific communication.
5. **ANCHORING PHENOMENON:** Did the final lesson successfully close the loop on the Lake Victoria fisherman scenario? Could learners articulate, in their own words, why sound bends around a headland and why water waves slow in a shallow bay? If not, the phenomenon was used as a contextual wrapper but not as a true explanatory target — revise Lesson 1's phenomenon introduction to make the 'puzzle' sharper and more personal for learners next year.

6. PERSONAL GROWTH: What did you learn about your learners' wave understanding today that you did not know before? Note one specific misconception that surfaced (e.g., confusing diffraction with reflection, or believing frequency changes on refraction) and plan how to address it proactively in the next cohort's Lesson 3 or 4.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	In this unit, students observed: sound reflecting off the rocky headland at Lake Victoria (echo experiment), water waves refracting as they crossed from deep open lake into the shallow Kisumu bay (wavefront diagrams and simulation), sound waves diffracting around a corner (classroom experiment with a sound source behind a barrier), and the fading of mobile-phone signals in Rift Valley valleys behind hills (case study and discussion).
What did I learn?	Students learned: the wave equation $v = f\lambda$ and how to use it to calculate speed, frequency, and wavelength in Kenyan contexts (Nairobi FM radio, Mombasa sonar, Kisumu water waves); the law of reflection and the echo-distance formula $d = (v \times t) / 2$; that refraction changes wave speed and direction but NOT frequency, and that wavelength shifts proportionally with speed; that diffraction spreads waves around obstacles and through gaps, and is greatest when wavelength $\approx$ gap size; that waves carry energy and can damage coastal communities (storm surges, coastal erosion), but that this impact can be mitigated through mangrove restoration, wave-energy barriers, and early-warning sonar systems.
How does this explain the phenomenon?	Students can now explain the full anchoring phenomenon: the Kisumu fisherman hears thunder from behind the cliff because sound waves DIFFRACT around the rocky headland — the low-frequency rumble has a long wavelength comparable to the headland gap, so it spreads into the cove. The water waves change direction at the edge of the shallow bay because of REFRACTION — the wave speed decreases in shallow water, bending the wavefronts toward the normal, steering wave energy into the protected cove even without a straight-line path from the storm. The mobile-phone signal that fades behind a Rift Valley hill behaves the same way as the sound around the headland — microwaves DIFFRACT around the hill, though imperfectly, which is why engineers place repeater towers on hilltops to boost the diffracted signal into valleys. Wave science therefore directly answers the driving question: understanding reflection, refraction, and diffraction — and the mathematics linking them through $v = f\lambda$ — equips communities, engineers, and scientists to protect lives, design better communications, and make sense of the energy-filled world around them.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.